
**Linear vs. Non-Linear Return Forecasting: A Dual
Evaluation of Predictive Accuracy and Economic
Value**

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Abstract

This paper reassesses the incremental value of non-linearity in equity return forecasting. Existing machine-learning evidence documents that flexible non-linear models outperform simple linear benchmarks, but it remains unclear whether these gains reflect non-linear structure or improved processing of high-dimensional firm characteristics. Using 3.4 million U.S. stock-month observations and more than 150 firm-level signals, the analysis compares non-linear specifications with linear models explicitly designed to handle high-dimensionality through dimension reduction. *Statistically, dimension-reduced linear models and non-linear specifications deliver comparable out-of-sample R^2 , with adjusted Diebold-Mariano tests showing no statistically significant difference after Bonferroni correction. Economically, non-linear models generate stronger long-short portfolio performance under value-weighting, while linear models remain competitive under an equal-weighting implementation once realistic transaction costs are incorporated.* Firm-level characteristics retain significant predictive content, concentrated in low risk, quality, momentum, value and profitability based signals, although their relative importance varies substantially over time. Overall, the results show that model complexity does not uniformly translate into superior forecasting performance and that dimension-reduced linear models remain essential benchmarks for evaluating the incremental value of non-linearity itself.

Keywords: Equity risk premium, machine learning, econometric modelling, portfolio construction, factor investing

1. Introduction and Motivation

A central claim in the machine-learning asset-pricing literature is that non-linear models improve return forecasts relative to simple linear benchmarks (e.g., Gu et al., 2020; Drobetz & Otto, 2021; Leung et al., 2021). However, such comparisons do not isolate non-linearity, since flexible non-linear models also regularise, aggregate and transform high-dimensional predictors. This paper therefore asks whether non-linear specifications retain incremental statistical and economic value once linear benchmarks are augmented with dimension reduction.

The main contribution is a stricter test of non-linearity, comparing non-linear models not with simple OLS but with dimension-reduced linear benchmarks. This isolates whether interactions and threshold effects add value beyond characteristic compression. The central hypothesis is therefore: *dimension-reduced linear models perform comparably to non-linear specifications, both in terms of statistical accuracy and economic value*

The empirical setting further contributes to this test. The analysis uses the Jensen et al. (2023) characteristics library, covering more than 150 firm-level signals and evaluates U.S. stock-month observations through 2024. Forecasting performance is assessed using out-of-sample R^2 , while economic value is evaluated through long-short decile portfolios that incorporate realistic transaction costs. Secondary analyses examine time-varying characteristic importance and reassess the predictive content of firm-level characteristics in light of the sceptical evidence in Cakici et al. (2024).

The main results are nuanced. Statistically, the evidence does not reveal a clear performance gap between dimension reduced linear models and non-linear specifications. Economically, non-linear models perform better mainly in value-weighted portfolios, while dimension-reduced linear models remain competitive under equal weighting. Across models and weighting schemes, portfolio performance is primarily driven by the long leg. Focusing on the characteristic-importance analysis, it further shows that firm-level signals retain meaningful predictive content. Predictive power is concentrated in low risk, quality, momentum, value and profitability signals, with relative importance that varies greatly over time. These patterns suggest that return predictability is concentrated in a limited set of economically interpretable dimensions (Feng et al., 2020) and that the sources of predictability are state-dependent.

The findings imply that non-linear complexity is not universally warranted. Dimension-reduced linear models offer a more transparent and easier-to-maintain alternative when their forecasts are statistically comparable. At the same time, non-linear models can add value in specific implementations, particularly value-weighted strategies. The economic value of machine-learning complexity therefore depends not only on predictive accuracy, but also on the portfolio construction rule through which forecasts are implemented.

The remainder of the paper proceeds as follows. Section 2 describes the forecasting models, evaluation metrics, data and pre-processing framework. Section 3 presents the main empirical results, including the comparative model analysis and characteristic importance evidence. Section 4 concludes by summarising the key findings, implications and limitations.

2. Methodology

2.1. Modelling Framework

All models share the same overarching structure: they attempt to estimate the conditional expected excess return of stock i at time $t + 1$:

$$r_{i,t+1} = \mathbb{E}_t(r_{i,t+1}) + \varepsilon_{i,t+1}, \quad \mathbb{E}_t(r_{i,t+1}) = g^*(z_{i,t}), \quad (1)$$

where $\mathbb{E}_t(r_{i,t+1})$ denotes the conditional expected excess return of stock i given information available at time t , $\varepsilon_{i,t+1}$ is the forecast error while $z_{i,t}$ is the P -dimensional vector of rank-transformed firm characteristics and $g^*(\cdot)$ is the model-specific mapping.

Stocks are indexed by $i = 1, \dots, N_t$ and months by $t = 1, \dots, T$. Models are estimated at the firm-month level, treating each observation independently conditional on its predictors, with the following models included:

Table 1: Included Models

Model Class	Specification
Linear	OLS, OLS-FF3, PCR, IPCA, PLS
Non-linear	GBRT, RF, FFNN
Ensemble	Equal-weighted linear ensemble, equal-weighted non-linear ensemble

Note: This table reports the set of forecasting models included in the empirical analysis. The model selection reflects the paper's focus on dimension reduction methods while preserving a balanced comparison between linear and non-linear specifications.

Penalised linear models (LASSO/Ridge) are excluded because shrinkage is suboptimal when predictors are highly correlated, as is typical for stock characteristics (Gu et al., 2020). Appendix A provides a complete breakdown of the models included in this paper.

2.2. Statistical Evaluation and Variable Importance

Forecasting performance is assessed using a unified framework that combines statistical accuracy, pairwise model comparison and characteristic-importance attribution.

Out-of-sample R^2 : The accuracy metric is computed over the test set $\mathcal{T}_3$. Following Gu et al. (2020), the denominator uses raw squared excess returns rather than demeaned values, benchmarking predictions against zero rather than the historical mean. This imposes a stricter benchmark and reduces the risk of overstating model accuracy:

$$R_{\text{oos}}^2 = 1 - \frac{\sum_{(i,t) \in \mathcal{T}_3} (r_{i,t+1} - \hat{r}_{i,t+1})^2}{\sum_{(i,t) \in \mathcal{T}_3} r_{i,t+1}^2}, \quad (2)$$

where $\hat{r}_{i,t+1}$ is the predicted excess return for stock i at time $t+1$ by the specified model.

Pairwise model comparison: Pairwise differences in out-of-sample R^2 are tested using the adjusted Diebold-Mariano (DM) statistic (Diebold & Mariano, 1995) specified in Gu et al. (2020). Prediction errors are first averaged cross-sectionally at each time step to produce a single time series of error differences, which satisfies the regularity conditions for asymptotic normality even under strong cross-sectional correlation. The test statistic is given by:

$$DM_{12} = \frac{\bar{d}_{12}}{\hat{\sigma}_{\bar{d}_{12}}}, \quad d_{12,t+1} = \frac{1}{n_{3,t+1}} \sum_{i=1}^{n_{3,t+1}} \left[(\hat{e}_{i,t+1}^{(1)})^2 - (\hat{e}_{i,t+1}^{(2)})^2 \right], \quad (3)$$

where $\hat{e}_{i,t+1}^{(k)}$ is the prediction error of model k for stock i at time t , $n_{3,t+1}$ represents the number of stocks in the testing sample (year $t+1$) and $\hat{\sigma}_{\bar{d}_{12}}$ is the Newey-West standard error computed with four lags (Newey & West, 1987; Drobetz & Otto, 2021). Statistical significance of out-of-sample R^2 relative to the zero benchmark is assessed separately via the one-sided Clark-West (CW) test (Clark & West, 2007):

$$H_0 : R_{\text{oos}}^2 \leq 0 \quad \text{versus} \quad H_A : R_{\text{oos}}^2 > 0. \quad (4)$$

To control for Type I errors under simultaneous testing, both the DM and CW statistics are adjusted with the Bonferroni correction, which divides the significance level by the number of simultaneously considered comparisons (Bonferroni, 1936).

Variable importance: To identify which firm characteristics drive predictive performance, variable importance VI_j is defined as the reduction in out-of-sample R^2 when all values of characteristic j are set to zero while all other model parameters are held fixed (Kelly et al., 2019; Gu et al., 2020):

$$VI_j = R_{\text{full}}^2 - R_{\text{no-}j}^2, \quad (5)$$

where R_{full}^2 denotes the predictive R^2 obtained from the complete model and $R_{\text{no-}j}^2$ represents the R^2 after excluding feature j . Therefore, a positive VI_j indicates that characteristic j contributes incremental predictive power, while a $VI_j \approx 0$ indicates redundancy given the remaining signals. The measure is computed at each re-estimation window, allowing the time-varying importance of each signal to be tracked.

2.3. Data and Pre-Processing

The primary dataset is the Jensen et al. (2023) Global Factor Data library, accessed via WRDS. Characteristics are rank-transformed and mapped to the interval $[-1, 1]$ on a cross-sectional, period-by-period basis following Freyberger et al. (2020); see Appendix B. Consequently, missing values are replaced with the cross-sectional median or zero if the

median is undefined. Industry effects are removed via a within-industry demeaning using the first two SIC digits as industry dummies (Gu et al., 2020). The dataset comprises roughly 3.4 million observations across more than 27 000 unique firms, with an average of more than 5 000¹ cross-sectional observations a month.

To avoid forward-looking bias, all firm-level characteristics from Jensen et al. (2023) are lagged by one month, in accordance with the dataset documentation.

2.4. Sample Splitting and Hyperparameter Tuning

The models are refitted once per year. Each time the model is refitted, the training window expands by one year and the 12-year validation sample rolls forward accordingly to include the most recent 12 months. Time window has been selected to include recent observations, while maintaining the same relative split structure as Gu et al. (2020), ensuring enough initial data to get reliable estimates at the start of the evaluation period and an out-of-sample period that is long enough to be representative (Welch & Goyal, 2008).

The hyperparameters are tuned using the train-validation-test structure. This procedure reduces the risk of overfitting to specific iterations or market regimes and helps ensure that the reported results reflect genuine out-of-sample predictive performance.

Table 2: Sample Split Scheme

Sub-Sample	Period	Purpose
Training	1970-1986 (17 years)	Model estimation
Validation	1987-1998 (12 years, rolling)	Hyperparameter tuning
Test	1999-2024 (26 years)	Out-of-sample evaluation

Note: This table reports the sample split used in the empirical analysis. The training sample is used for model estimation, the validation sample for hyperparameter tuning and the test sample for out-of-sample performance evaluation.

See Appendix C for a visual representation of the data classification timeline and Appendix D for the estimated models and their parameters.

2.5. Portfolio Evaluation

Motivated by evidence that statistical forecast accuracy need not translate directly into economic profitability (Leitch & Tanner, 1991), the analysis complements the statistical evaluation with a portfolio based assessment of economic value. Specifically, the forecasts from each model are evaluated through portfolio sorts, allowing predictive performance to be assessed in terms of realised investment profitability rather than statistical fit alone. At the end of each month, stocks are sorted into ten decile portfolios based on their predicted excess returns. Long-short portfolios go long the top decile and short the bottom decile, with both equal-weighted and value-weighted portfolios constructed. The primary performance metrics are the Sharpe ratio (SR) (Sharpe, 1994) and the Probabilistic Sharpe Ratio (PSR) (Bailey & López de Prado, 2012). The SR provides the industry standard measure of performance, while the PSR accounts for estimation uncertainty and departures from idealised return assumptions that may otherwise obscure downside risk. They are defined as:

$$SR = \frac{\bar{r}}{\hat{\sigma}_r}, \quad PSR = \Phi \left(\frac{(SR - SR^*)\sqrt{n-1}}{\sqrt{1 + \frac{1}{2}SR^2 - \hat{\gamma}_3SR + \frac{\hat{\gamma}_4-3}{4}SR^2}} \right), \quad (6)$$

where $\bar{r}$ denotes the sample mean return, $\hat{\sigma}_r$ the sample standard deviation, SR^* the market SR, n the number of observations, $\hat{\gamma}_3$ sample skewness, $\hat{\gamma}_4$ sample kurtosis and $\Phi(\cdot)$ the standard normal cumulative distribution function.

Risk-adjusted performance is assessed using Capital Asset Pricing Model (CAPM) alpha (Sharpe, 1964; Lintner, 1965) and Fama-French 5-Factor (FF5) alpha (Fama & French, 2015), while macroeconomic co-movement is examined using the variables of Welch and Goyal (2008).

3. Empirical Results

3.1. Statistical Forecasting Performance

Table 3 reports the main out-of-sample R^2 results. Among the linear specifications, dimension reduction improves predictive accuracy relative to simple benchmarks. In particular, PLS delivers a higher out-of-sample R^2 than the OLS-based models, suggesting that compression of the characteristic space helps isolate relevant information. Across individual models, RF and FFNN2 achieve the strongest predictive performance, suggesting that non-linearities and interaction effects may provide incremental forecasting gains. At the same time, the ensemble models outperform their constituent models, consistent with Bali et al. (2023), who show that aggregating forecasts can improve predictive performance.

¹Following Gu et al. (2020) the firms are not filtered by share price, share codes or industry, thereby creating the broadest possible dataset for the analysis. There are three primary reasons for this, first these commonly applied filters exclude several constituents of the S&P 500 index, making their removal problematic for any asset-pricing analysis that aims to capture the full cross-section. Second, retaining the full sample mitigates concerns regarding sample-selection and data snooping biases (e.g., Lo & MacKinlay, 1990). Third, a larger sample reduces the risk of overfitting by increasing the ratio of observations to estimated parameters.

Table 3: Monthly Out-of-Sample R^2 (%)

Panel A: Linear Models							
	Simple Linear		Dimension Reduction				Ensemble
	OLS	OLS-FF3	PCR	IPCA	PLS		Linear-En
R^2_{OOS} (%)	0.20	0.12	0.08	0.10	0.26		0.37
CW statistic	(5.65)*	(2.40)	(4.51)*	(4.85)*	(5.17)*		(5.45)*
Panel B: Non-Linear Models							
	Tree-Based		Deep Learning				
	GBRT	RF	FFNN1	FFNN2	FFNN3	FFNN4	FFNN5
R^2_{OOS} (%)	0.29	0.36	0.31	0.36	0.26	0.30	0.25
CW statistic	(5.16)*	(4.60)*	(6.23)*	(6.50)*	(5.65)*	(5.74)*	(4.65)*
							Non-Linear-En
							0.43
							(5.21)*

Note: This table reports monthly out-of-sample stock-level R^2 values for each predictive model. Values in parentheses are the Clark and West (2007) statistics. Bold font indicates significance at the one-sided 5% level, while an asterisk indicates significance after Bonferroni adjustment. Newey-West standard errors are computed with four lags, following Drobetz and Otto (2021).

Another key implication of Table 3 is that the absolute magnitude of the models' forecasting ability is broadly comparable to that reported by Gu et al. (2020). This similarity is notable because the present analysis uses a more recent sample period and excludes macroeconomic interaction terms, thereby reducing the dimensionality of the predictor space and limiting the risk of overfitting (Rasekhschaffe & Jones, 2019). The results therefore suggest that the main predictive content of firm-level characteristics remains robust in an updated empirical setting, even when the model specification is made more parsimonious.

The predictive content of firm-level characteristics is further supported by the CW statistics reported in Table 3. With the exception of the OLS-FF3 specification, the reported test statistics are significant even after applying the stringent Bonferroni correction. This evidence indicates that characteristic-based forecasts deliver statistically meaningful improvements relative to the zero-return benchmark. This finding contrasts with Cakici et al. (2024), who report limited evidence of significant out-of-sample gains from firm-level characteristics. A likely explanation is the difference in benchmark construction. The present analysis follows the forecasting framework of Gu et al. (2020), whereas Cakici et al. (2024) evaluate predictive performance relative to the benchmark proposed by Welch and Goyal (2008). Since out-of-sample R^2 and CW statistics are inherently benchmark dependent, differences in the reference forecast can materially affect statistical inference.

Consistent with the paper's central hypothesis, Table 4 shows that the non-linear specifications do *not* statistically outperform their dimension-reduced linear counterparts once the Bonferroni correction is applied. However, in line with Gu et al. (2020) and Drobetz and Otto (2021), the non-linear models significantly improve upon the simple OLS specification. Beyond this case, the adjusted DM statistics provide little evidence that the greater functional flexibility of non-linear methods translates into a statistically significant gain in out-of-sample predictive accuracy.

This result suggests that the forecasting gains previously attributed to non-linear machine-learning models (e.g., Moritz & Zimmermann, 2016; Messmer, 2017; Leung et al., 2021) may not arise from non-linearity itself. If non-linear interactions and threshold effects were the primary source of superior predictive performance, these models would be expected to retain a clear advantage over augmented linear specifications. Instead, the evidence indicates that much of the apparent benefit of machine-learning may reflect the ability to regularise, compress or aggregate a large set of firm-level predictors. Once linear models are equipped with dimension-reduction methods such as PCR, IPCA and PLS, the incremental statistical value of non-linearity becomes limited.

Table 4: Adjusted DM Test Comparison of Out-of-Sample Predictive Performance

	OLS-FF3	PCR	IPCA	PLS	Linear-En	GBRT	RF	FFNN1	FFNN2	FFNN3	FFNN4	FFNN5	Non-Linear-En
OLS	-0.26	-0.66	-0.92	1.79	2.08	0.67	3.92*	0.88	1.74	0.38	1.56	1.05	5.07*
OLS-FF3		-0.08	-0.09	0.49	1.09	0.48	0.93	0.48	0.68	0.34	0.56	0.50	1.02
PCR			-0.02	1.26	2.51	1.05	1.88	0.80	1.10	0.62	0.99	1.01	1.80
IPCA				1.70	3.26*	1.27	2.24	1.03	1.43	0.80	1.26	1.19	2.32
PLS					1.42	0.09	2.45	0.05	0.63	-0.41	0.24	-0.04	2.49
Linear-En						-0.74	0.09	-0.73	-0.34	-1.08	-0.80	-1.11	0.43
GBRT							0.96	-0.03	0.36	-0.34	0.05	-0.09	1.29
RF								-1.07	-0.59	-1.66	-1.41	-2.39	0.68
FFNN1									2.08	-1.34	0.23	-0.08	2.72
FFNN2										-0.90	-0.68	1.91	
FFNN3										-2.82	1.21	0.42	4.27*
FFNN4												-0.33	3.88*
FFNN5													2.60

Note: Adjusted DM test statistic comparing out-of-sample stock-level prediction performance. Positive values indicate the column model outperforms the row model. Bold font indicates conventional significance at the one-sided 5% level while an asterisk (*) indicates significance under Bonferroni adjustment, applying four lags in computing the Newey-West standard errors, following Drobetz and Otto (2021).

3.2. Which Characteristics Matter?

Given the high dimensionality of the characteristic set, individual predictors are grouped into economically interpretable signal themes following the classification in Jensen et al. (2023). Across model specifications, five themes account for the largest share of predictive information: low risk, quality, momentum, value and profitability. This concentration suggests that return predictability is not evenly distributed across the characteristic space, but instead resides in a limited set of economically meaningful dimensions, consistent with Feng et al. (2020).

The relative importance of these themes, however, varies substantially over time. A particularly pronounced shift occurs around the 2008 Global Financial Crisis, when low risk signals become markedly more important while momentum related signals decline sharply. This pattern is in line with the evidence in Dang and Nguyen (2020), who show that ex ante proxies for stock-level liquidity risk help explain realised equity returns during the crisis. It is also consistent with the momentum-crash mechanism documented by Daniel and Moskowitz (2016), whereby momentum strategies perform poorly during periods of abrupt market reversals and heightened funding or liquidity stress. Taken together, the results indicate that the economic sources of predictability are state-dependent.

A second notable pattern is the difference in thematic stability across model classes. Linear specifications exhibit substantially lower variation in signal theme importance across time than their non-linear counterparts. This difference can be attributed to the distinct ways in which the models process high-dimensional characteristic information. Dimension reduction methods project individual predictors onto a smaller set of latent components, thereby smoothing idiosyncratic fluctuations in individual signals and producing more stable importance profiles. By contrast, non-linear specifications are more sensitive to changes in the predictive environment. Tree-based and neural network models can reallocate importance sharply across characteristics as the relevant splits, interactions or learned representations change across market states. The model complexity evidence supports this interpretation. Non-linear models tend to retain and use a large share of the original characteristic set, whereas the dimension-reduced specifications adjust the number of retained components. This also implies more volatile importance estimates for the flexible non-linear models. Appendix F reports an in-depth overview of characteristic importance, economic intuition and model complexity.

3.3. Economic Value and Portfolio Implications

3.3.1 Portfolio Performance

To assess practical implementation, the paper reports maximum drawdown, average monthly turnover, the most extreme one-month loss and transaction costs. Maximum drawdown and turnover are defined as:

$$\text{MaxDD} = \max_{0 \leq t_1 \leq t_2 \leq T} (Y_{t_1} - Y_{t_2}), \quad \text{Turnover} = \frac{1}{T} \sum_{t=1}^T \sum_i |w_{i,t+1} - \tilde{w}_{i,t+1}|. \quad (7)$$

Here Y_t denotes the cumulative log return from inception to time t and $\tilde{w}_{i,t+1}$ is the pre-rebalancing buy-and-hold weight implied by the previous portfolio weights and realised returns. For long-short portfolios, the long and short legs are drifted and normalised separately before computing turnover. Maximum drawdown measure captures the largest observed difference in portfolio value from a peak to the subsequent low point while turnover measures the average trading required to rebalance the post-return portfolio weights back to their target allocation. Transaction costs are estimated using two approaches: the methodology of Bollerslev et al. (2018), Koijen et al. (2018), and Drobetz and Otto (2021) and Interactive Brokers' commission schedule, defined as:

$$\text{TC}_t^{(1)} = \text{OT}_t \times \text{HS}_t, \quad \text{TC}_t^{(2)} = 0.005 \times N_t^{\text{shares}}, \quad (8)$$

where:

$$\text{OT}_t = \frac{1}{2} \text{Turnover}, \quad \text{HS}_t^{\text{EW}} = \frac{1}{n} \sum_{i=1}^n \frac{1}{2} \frac{\text{Ask}_{i,t} - \text{Bid}_{i,t}}{\frac{1}{2}(\text{Ask}_{i,t} + \text{Bid}_{i,t})}, \quad \text{HS}_t^{\text{VW}} = \sum_{i=1}^n w_{i,t} \frac{1}{2} \frac{\text{Ask}_{i,t} - \text{Bid}_{i,t}}{\frac{1}{2}(\text{Ask}_{i,t} + \text{Bid}_{i,t})}. \quad (9)$$

Here, OT_t is the one-sided turnover, HS_t is the respective halfspread, $\text{TC}_t^{(1)}$ follows Bollerslev et al. (2018), Koijen et al. (2018), and Drobetz and Otto (2021) and $\text{TC}_t^{(2)}$ follows the Interactive Brokers' commission schedule. Specifically, $\text{TC}_t^{(2)}$ is applied only when $\text{TC}_t^{(1)}$ cannot be computed, ensuring that portfolio performance is not overstated when transaction cost data are unavailable.

After incorporating realistic transaction costs, Table 5 reports the economic performance of the long-short portfolios. Under equal-weighting, dimension-reduced linear models remain competitive, as indicated by both the SR and PSR. Under value-weighting, however, the performance gap between linear and non-linear specifications becomes more pronounced, suggesting that the economic value of model flexibility is stronger when portfolios are tilted towards larger and more liquid firms. In the value-weighted portfolios, statistically meaningful performance is concentrated among the non-linear specifications. Only these strategies generate SRs and PSRs that remain significant relative to the market. This pattern suggests that, when portfolio weights are tilted towards larger and more liquid firms, the economic gains from return forecasting are primarily delivered by models with greater flexibility. However, despite this performance, both the SR and

PSR remain lower under value-weighting than under equal-weighting. A potential explanation for this divergence between portfolio performance is because the statistical loss functions are designed to minimise equal-weighted forecast errors, creating a structural mismatch when portfolios are value-weighted (Gu et al., 2020). These findings are broadly consistent with the existing asset-pricing literature, which documents stronger portfolio performance for non-linear models and illustrates the divergence between equal-weighted and value-weighted implementations (e.g., Gu et al., 2020; Drobetz & Otto, 2021).

Table 5: Long-Short Portfolio Performance for the Out-of-Sample Period

Model	Equal-Weighted				Value-Weighted			
	Pred (%)	Real (%)	SR	PSR	Pred (%)	Real (%)	SR	PSR
Market (benchmark)	-	0.63	0.14	-	-	0.63	0.14	-
OLS	5.37	2.00	0.32	0.99	4.91	0.93	0.16	0.64
OLS-FF3	0.00	-0.28	-0.11	0.00	0.00	0.20	0.05	0.06
PCR	2.47	1.45	0.20	0.83	2.41	0.75	0.09	0.21
IPCA	4.01	1.45	0.21	0.92	3.76	0.41	0.06	0.09
PLS	4.11	1.81	0.30	0.99	3.88	0.89	0.13	0.47
Linear-En	2.83	2.08	0.35	0.99	2.64	0.10	0.16	0.62
GBRT	5.70	1.86	0.34	0.99	4.85	1.68	0.23	0.95
RF	2.92	1.96	0.31	0.99	2.23	1.43	0.17	0.74
FFNN1	7.13	2.33	0.38	0.99	5.70	1.55	0.21	0.93
FFNN2	6.89	2.25	0.36	0.99	5.47	1.83	0.25	0.99
FFNN3	6.55	2.14	0.35	0.99	5.40	1.42	0.20	0.88
FFNN4	5.80	2.11	0.34	0.99	4.91	1.44	0.20	0.88
FFNN5	4.00	2.04	0.33	0.99	3.46	1.57	0.23	0.97
Non-Linear-En	5.22	2.34	0.37	0.99	4.27	1.66	0.22	0.94

Note: This table reports the performance of long-short decile portfolios over the 26-year out-of-sample test period. For each model, the table reports the average predicted return, the average net realised return, the SR and the PSR, separately for equal-weighted and value-weighted portfolio construction. Predicted and net realised returns are expressed in monthly percentages, while SRs are computed from net realised monthly portfolio returns. Bold SRs indicate portfolios that exceed the monthly SR of the market excess return over the same out-of-sample period, equal to 0.14. The PSRs report the probability that the estimated portfolio SR exceeds the market SR, with probabilities lying in the interval $[0,1]$. Likewise bold PSR values indicate statistical significance at the 5% level relative to this benchmark.

3.3.2 Risk-Adjusted Performance and Macro Co-Movement

Risk-adjusted performance is evaluated to determine whether observed return spreads reflect compensation for systematic risk², while macroeconomic co-movement assesses whether portfolio returns are primarily driven by macro factors or business-cycle fluctuations (Asness et al., 2013).

After controlling for systematic risk exposures through the CAPM and FF5 specifications, the economic advantage of non-linear models shown in Table 5 persists. Non-linear specifications continue to generate large and statistically significant alphas across weighting schemes, indicating that their performance is not merely compensation for standard factor exposures. By contrast, the alphas of the linear models are weaker under value-weighting but remain competitive under equal-weighting. This further suggests that, when portfolio weights are tilted towards larger and more liquid firms, the economic gains from return forecasting are primarily generated by models with greater flexibility.

Focusing on downside risk, maximum drawdowns and extreme one-month losses are generally more pronounced for the linear specifications. This pattern suggests that, although dimension-reduced linear models can remain competitive in average risk-adjusted performance, they may be less effective at managing adverse tail realisations in long-short portfolio returns. A possible explanation is that linear models impose a smoother and more persistent mapping from characteristics to expected returns. This structure can make portfolio positions slower to adjust when predictive relations deteriorate abruptly, a pattern consistent with their lower absolute volatility in signal importance over time. By contrast, non-linear models appear more responsive to short-lived shifts in the predictive environment. This greater flexibility is reflected in higher turnover and reinforced by more volatile signal importance profiles, indicating more frequent portfolio reallocation as these models update the cross-sectional ranking of expected returns. The higher turnover may also reflect the greater importance assigned by non-linear specifications to price-trend characteristics, whose predictive content can vary sharply over time and whose portfolio implications may involve rapid position changes (Gu et al., 2020). Although momentum-related characteristics are among the most important predictors across model classes, they are particularly influential in the non-linear specifications.

Using the macroeconomic variables of Welch and Goyal (2008), the evidence suggests that long-short portfolio performance is only weakly related to aggregate business-cycle conditions. Most macroeconomic state variables do not statistically explain portfolio returns. This pattern indicates that the profitability of the model strategies is not primarily driven by conventional time-series variation in expected returns associated with the macroeconomy. This finding is consistent with the broader asset-pricing literature, which documents the limited and unstable forecasting power of macroeconomic variables. In particular, Welch and Goyal (2008) show that many standard macroeconomic predictors perform poorly out-of-sample, while Rapach and Zhou (2013) emphasise that the predictive content of macroeconomic variables are often subject to samples and model specifications. Moving on to the estimated alphas, statistically significant intercepts are concentrated in the equal-weighted portfolios, whereas the macro-adjusted alphas are weaker under value-

²In a linear factor model, the tangency portfolio of the factors themselves represents the maximum SR portfolio in the economy. If any portfolio delivers SR beyond the factor tangency portfolio it possesses alpha with respect to that model (Gu et al., 2020)

weighting. This distinction is in line with Table 5 and the argument that once portfolio returns are tilted towards larger and more liquid firms, the abnormal performance becomes less pronounced.

3.3.3 Cumulative Performance

The cumulative portfolio performance reinforces the findings from Table 5: equal-weighted long-short portfolios generally outperform the market, whereas value-weighted portfolios show weaker performance. Across both weighting schemes, non-linear models tend to outperform their linear counterparts.

A consistent finding across models and weighting schemes is that portfolio performance is predominantly driven by the *long leg*: the models are much better at identifying future outperformers than future underperformers, a pattern consistent with Gu et al. (2020).

Appendix G provides the description of the macroeconomic variables, risk-adjusted performance, macroeconomic co-movement and cumulative portfolio performance.

3.4. Robustness

The results reported in Table 4 are robust to a broad set of alternative empirical specifications, including winsorisation, alternative characteristic databases, market-capitalisation subsamples, alternative industry classifications and alternative sample-splitting schemes. Across these robustness exercises, dimension-reduced linear models and non-linear specifications remain statistically indistinguishable in terms of out-of-sample R^2 . This evidence reinforces the interpretation that previously documented gains from non-linear machine-learning models may partly reflect comparisons against simple linear benchmarks, rather than a distinct incremental contribution from non-linearity itself. The CW statistics are similarly stable across robustness settings, indicating that firm-level characteristics retain significant predictive content even when the empirical design is varied.

The economic results are broadly consistent with the baseline findings: non-linear models continue to deliver stronger portfolio performance than linear specifications, suggesting that their economic value is more persistent than their incremental statistical forecasting advantage. The large-capitalisation subsample is the main exception. If the value-weighted gains reflected superior signal extraction among large and liquid firms, as argued in Section 3.3.1, non-linear models should dominate precisely in that universe. They do not. Linear specifications perform better, indicating that the value-weighted advantage of non-linear models cannot be attributed simply to large-capitalisation signal extraction.

This distinction underscores the central message of the paper: model complexity does not necessarily deliver statistically superior stock-level forecasts, but it can still improve realised portfolio outcomes once forecasts are translated into investment decisions, consistent with the distinction between statistical forecast accuracy and economic value documented by Leitch and Tanner (1991).

Appendix H reports an overview of the distinct robustness settings with the accompanying DM statistic and portfolio performance.

4. Conclusion

This paper examines whether machine-learning complexity delivers incremental value in equity return forecasting. The evidence points towards a nuanced answer. Once linear models are augmented with dimension reduction, non-linear specifications *no* longer exhibit a statistically robust forecasting advantage. Across 55 years of U.S. equity data, more than 150 firm-level characteristics and extensive robustness tests, dimension-reduced linear and non-linear methods are largely indistinguishable in out-of-sample predictive accuracy.

The economic evidence, however, shows that non-linearity can still matter once forecasts are translated into portfolios. Non-linear models generate stronger long-short performance in several portfolio implementations. The value of machine-learning complexity therefore depends not only on statistical forecast accuracy, but also on the investment rule through which forecasts are implemented.

Several limitations and areas for future research remain. First, the analysis is limited to U.S. equities, external validity should be tested in international markets. Second, the portfolio tests include transaction costs but do not fully account for market impact, short-borrow costs, financing constraints or capacity limits. Third, although an explanation for the divergence between equal- and value-weighted portfolios is given, it has not been formally tested. Finally, the divergence between statistically similar out-of-sample R^2 and different portfolio outcomes warrants further analysis of rank correlations, decile hit rates and tail-specific forecast accuracy.

These findings reframe the model selection problem in quantitative asset management. The relevant question is not simply whether a model is linear or non-linear, but whether its complexity adds value beyond disciplined feature processing, robust validation and economically realistic portfolio construction. Dimension-reduced linear models remain essential benchmarks: they are transparent, simple and statistically competitive. Non-linear models are valuable when their flexibility survives the transition from prediction to implementation. In short, complexity is not a substitute for economic discipline; it is only useful when it improves realised investment outcomes.

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Appendix

A. Models Employed in the Research

A.1. Ordinary Least Squares

For the OLS models, two simple linear approaches are constructed: a pooled OLS specification using the full set of firm-level characteristics and a Fama-French 3-Factor (FF3) model (Fama & French, 1993). The pooled OLS has been widely used as a benchmark in the return-forecasting literature (e.g., Messmer, 2017; Gu et al., 2020), while the FF3 specification provides a factor-based benchmark that improves substantially on the traditional CAPM (Gu et al., 2020). The two linear specifications are defined as:

$$g(z_{i,t}; \theta) = z'_{i,t} \theta, \quad g(f_t; \beta_i) = f'_t \beta_i. \quad (10)$$

Here, $z_{i,t}$ denotes the vector of firm-level characteristics for stock i at time t and θ is the corresponding pooled coefficient vector. In the FF3 specification, f_t is a 3×1 vector containing the market, size and value factors, while β_i is a 3×1 vector of firm-specific factor sensitivities.

The simple linear approach is extended by incorporating a robust objective function to address the pronounced heavy tails often observed in financial and economic data, thereby improving the stability of forecasts in the presence of large outliers through the application of the Huber loss function.³ The Huber loss is a hybrid between the squared and absolute loss functions; it behaves like the squared loss for small errors and switches to an absolute loss for large errors, reducing the influence of extreme observations.

A.1.1 Huber Loss Function

The Huber loss function (Huber, 1992) is an objective function that provides a compromise between the squared loss and the absolute loss, making it robust under heavy-tailed distributions. Formally, the Huber loss objective is defined as:

$$\mathcal{L}_H(\theta) = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T H(r_{i,t+1} - g(z_{i,t}; \theta), \xi), \quad (11)$$

where $r_{i,t+1}$ denotes the realised excess return for asset i at time $t + 1$, $g(z_{i,t}; \theta)$ is the model-predicted return based on covariates $z_{i,t}$ and parameters θ while ξ is a tuning parameter threshold controlling the transition between quadratic and linear loss. Where:

$$H(x; \xi) = \begin{cases} x^2, & \text{if } |x| \leq \xi, \\ 2\xi|x| - \xi^2, & \text{if } |x| > \xi. \end{cases} \quad (12)$$

The Huber loss function exhibits a quadratic behaviour for small residuals ($|x| \leq \xi$) and transitions to a linear form for large residuals ($|x| > \xi$). To illustrate this, consider the case where $\xi = 1$ and there are two observations: $x_1 = 0.5$ and $x_2 = 1.5$. For x_1 , the loss is computed as the mean squared error: $x_1^2 = 0.25$. While for x_2 , the loss follows the linear loss, yielding $2 \cdot 1 \cdot x_2 - 1^2 = 2$. Under the conventional mean squared error framework, the corresponding value for x_2 would have been $x_2^2 = 2.25$.

The threshold parameter ξ defines the transition point between the quadratic and linear loss and is set adaptively from the data.

A.2. Dimension Reduction Linear Models

Dimension-reduction methods reduce the original P -dimensional predictor space to a lower-dimensional set of K components, where $K < P$. The general objective is to replace the full characteristics matrix with a smaller set of linear combinations that retain the information most relevant for forecasting returns. Let R denote the $NT \times 1$ vector of returns stacked across firms and time, while Z denotes the corresponding $NT \times P$ matrix of firm-level characteristics. The unrestricted linear forecasting model can be written as:

$$R = Z\theta + E, \quad (13)$$

where θ is a $P \times 1$ coefficient vector and E is the residual vector. Dimension reduction replaces Z with a transformed matrix of K components:

$$R = (Z\Omega_K)\theta_K + \tilde{E}, \quad (14)$$

where Ω_K is a $P \times K$ loading matrix that maps the original predictors into K lower-dimensional components, $Z\Omega_K$ denotes the transformed predictor matrix, θ_K contains the corresponding regression coefficients and $\tilde{E}$ is the prediction

³See Tukey (1960) and Huber (1964) for foundational work.

error. Principal Component Regression (PCR) and Partial Least Squares (PLS) share this general structure, but differ in how the loading matrix Ω_K is constructed.

A.2.1 Principal Component Regression

PCR constructs Ω_K using principal component analysis (PCA). It selects the linear combinations of predictors that explain the largest share of variation in the characteristics matrix Z . The intuition is that predictors can be dropped without a significant loss of information, as they can be reconstructed from the remaining predictors. PCR therefore exploits the correlation among variables, reducing the number of predictors while retaining as much of the information contained in the original data as possible (Stock & Watson, 2019). These components are then used as regressors in equation 14. In this way, PCR improves on OLS specifications by reducing multicollinearity and noise while preserving the dominant structure in the predictor space.

A.2.2 Partial Least Squares

PLS follows the same forecasting structure in equations 13 and 14, but constructs Ω_K differently. Unlike PCR, which selects components based on the variance of the predictors, PLS selects components that maximise covariance with the target variable R . In this way, PLS improves upon simple OLS by being more directly oriented toward prediction, since the extracted components are chosen not only to summarise the characteristics matrix, but also to be informative about future returns.⁴

A.2.3 Instrumented Principal Component Analysis

IPCA models the cross-section of returns by linking latent factor loadings to observable firm characteristics (Kelly et al., 2019). An intuitive way to understand IPCA is through the lens of characteristic-managed portfolios⁵: for each characteristic $l = 1, \dots, L$, the cross-section of returns is projected on lagged firm characteristics, producing a managed portfolio return $X_{l,t}$. Stacking these returns over time gives a $T \times L$ matrix X , from which IPCA extracts common latent factors:

$$\hat{F}_t = \hat{\Gamma}' X_t = (\hat{\Gamma}' Z_{t-1}) R_t. \quad (15)$$

Here, $X_t \in \mathbb{R}^{L \times 1}$ is the vector of managed portfolio returns in period t , $\hat{F}_t \in \mathbb{R}^{K \times 1}$ are the K latent factors, $\hat{\Gamma} \in \mathbb{R}^{L \times K}$ is the matrix that rotates the managed portfolios into the latent factor space, $Z_{t-1} \in \mathbb{R}^{N \times L}$ is the matrix of lagged firm characteristics and $R_t \in \mathbb{R}^{N \times 1}$ is the vector of realised excess returns.

This formulation shows that IPCA applies dimension reduction to characteristic-managed portfolio returns rather than directly to individual stock returns. Unlike PCR, which extracts components from the predictor matrix alone, IPCA estimates latent factors whose loadings vary with firm characteristics, making the model more interpretable in terms of time-varying risk exposures.

Kelly et al. (2019) distinguish between restricted and unrestricted IPCA. The restricted model sets $\Gamma_\alpha = 0$, so returns are explained only through characteristic-driven factor exposures, ruling out the possibility of anomalies (i.e., risk-free returns):

$$r_{i,t+1} = z'_{i,t} \Gamma_\beta f_{t+1} + \varepsilon_{i,t+1}, \quad (16)$$

where f_{t+1} represents latent factors and Γ_β maps characteristics to factor loadings. The unrestricted model allows $\Gamma_\alpha \neq 0$, introducing a characteristic-based intercept that captures predictable return variation not explained by latent factor exposure (i.e. exposure to systematic risk):

$$r_{i,t+1} = z'_{i,t} \Gamma_\alpha + z'_{i,t} \Gamma_\beta f_{t+1} + \varepsilon_{i,t+1}, \quad (17)$$

where $z'_{i,t} \Gamma_\alpha$ captures the characteristic-based intercept and $z'_{i,t} \Gamma_\beta f_{t+1}$ represents the systematic, risk-based component of returns.

This paper uses the restricted specification, as Kelly et al. (2019) show that the unrestricted model provides limited additional predictive gains once multiple common factors are included.

Following Kelly et al. (2019), the instrument vector separates each raw characteristic into a persistent and time-varying component. For characteristic $c_{i,t}$, the model includes its firm-specific time-series mean $\bar{c}_i$, its deviation $c_{i,t} - \bar{c}_i$ and a constant:

$$z_{i,t} = [1, \bar{c}_i, c_{i,t} - \bar{c}_i]. \quad (18)$$

For C raw characteristics, the instrument set contains $2C + 1$ variables. The model is estimated by Alternating Least Squares, iterating between latent factor realisations and the characteristic-based loading matrix.

⁴Prof. Dr. Maxim Ulrich made a valuable observation that the target variable in stock return prediction typically exhibits low variance. If one removes low variance components, they risk discarding the predictive signal. Since PLS maximizes the covariance between predictors and the target variable, it may underweight important low variance components, thereby reducing predictive accuracy. Despite these concerns PLS is included in the analysis, given its prominence as a linear dimension reduction method and the potential added contribution to the ensemble model.

⁵This interpretation was made during the Utah Winter Finance Conference in 2018 by Brian T. Kelly and closely follows the methodology.

In this way, IPCA improves upon the simple linear specification by incorporating time-varying parameters that capture the dynamic behaviour of characteristics. By linking factors to economic characteristics, it identifies time-varying risk exposures and predictive signals, resulting in a more robust and interpretable model for stock returns.

A.3. Tree-Based Models

Gradient boosting regression trees (GBRT) and Random Forest (RF) build on decision tree learners, which partition the target variable into binary groups with the highest possible intra-group homogeneity. At a fundamental level, trees start with the whole dataset, splitting it into smaller "branches" where observations behave similarly. Each new branch sorts the remaining data from the previous step, "growing" the tree as it progresses. Furthermore, they possess multiple desirable attributes for prediction models. They rely exclusively on the ordering of the predictive values and are therefore invariant to monotonic transformations of the covariates and can incorporate both categorical and numerical data within the same model. In addition, the structure of regression trees captures interactions and non-linearity. Despite these advantages, trees are prone to overfitting and must therefore be regularised (Wolff & Neugebauer, 2019; Gu et al., 2020). Formally, the forecast of a decision tree, $\mathcal{T}$, with K terminal nodes, can be written as:

$$g(z_{i,t}; \theta, K, L) = \sum_{k=1}^K \theta_k 1\{z_{i,t} \in C_k(L)\}, \quad (19)$$

where $C_k(L)$ denotes one of the K partitions of the data. Each partition is a product of up to L indicator functions of the predictors. The parameter k (denoted θ_k) are defined as the sample average of outcomes within the partition.

A.3.1 Boosting

The first tree regularisation approach is "boosting", which sequentially combines forecasts from deliberately over-simplified trees. These shallow trees are "weak learners" individually, but many such trees may as an ensemble produce a single "strong learner" (Friedman, 2001; Gu et al., 2020).

The procedure starts by fitting a shallow tree (e.g. depth of one). The next shallow tree, with the same depth, is used to fit the residuals of the first tree, to capture variations unexplained by the first tree. Predictions by the second tree are shrunk by the shrinkage rate, to prevent the ensemble model from overfitting the residuals. This procedure is repeated sequentially, at each new step b a shallow tree is fitted to the residuals of tree $b-1$. The residual forecast is then added to the ensemble model with the shrinkage weight. After B iterations, the ensemble model consists of many weak learners, with three tuning parameters, tree depth, shrinkage rate and iterations (Gu et al., 2020)

A.3.2 Random Forest

The second regulariser is RF, which likewise is an ensemble method. RF modifies Breiman (2001) bootstrapping aggregation method, or "bagging".

The approach starts by the baseline bagging procedure which draws B different bootstrap samples of the original dataset, fits deep trees independently and then averages their forecasts in an ensemble forecast. Dominant predictors are expected to be more prevalent split variables, especially at lower levels, resulting in substantial correlation among their final forecasts. Therefore, RF de-correlates trees by applying a dropout method, randomly dropping predictors at each branch so only a subset of randomly drawn predictors is considered. This in turn lowers the correlation by ensuring that early branches are forced to split on alternative predictors. Tuning parameters consist of tree depth, number of trees and number of predictors randomly considered as split variables (Wolff & Neugebauer, 2019; Gu et al., 2020).

A.4. Neural Networks

At a high level, a feedforward neural network (FFNN) maps the raw predictors into an outcome prediction through a sequence of layered transformations. The input layer contains the original predictors, while the hidden layers form non-linear interactions among those predictors and the output layer aggregates the learned representation into the outcome prediction (Rumelhart et al., 1986).

In total, five FFNN architectures are considered. Following the geometric pyramid rule of Masters (1993), the number of neurons is reduced progressively across successive hidden layers. The shallowest architecture, NN1, contains a single hidden layer with 32 neurons. NN2 contains two hidden layers with 32 and 16 neurons, while NN3 extends the structure to three hidden layers with 32, 16 and 8 neurons. NN4 adds a fourth hidden layer with 4 neurons and NN5 uses five hidden layers with 32, 16, 8, 4 and 2 neurons, respectively. Although deeper networks can achieve the same results as shallower networks with far fewer parameters (Eldan & Shamir, 2016), network depth must be balanced against the size of the sample. The analysis therefore considers a smaller set of five FFNN architectures controlling for the available data (Gu et al., 2020).

The next consideration is the activation function, which introduces non-linearity in the FFNN architecture. Specifically, it defines the output of a neuron given a set of inputs, thereby determining whether a particular neuron should be

"activated" based on its inputs. Following Gu et al. (2020), the FFNN specifications use the rectified linear unit (ReLU)⁶ activation function. In its simplest form, ReLU is applied after the weighted sum of the previous-layer inputs and the bias term have been computed. The ReLU activation is then applied by retaining positive values and setting non-positive values to zero. The resulting activation is passed forward as an input to the next layer, allowing the network to build non-linear interactions across successive layers.

In its general form, the FFNN can be written as:

$$x_{i,t,k}^{(l)} = \text{ReLU} \left(\left(x_{i,t}^{(l-1)} \right)' \theta_k^{(l-1)} \right), \quad k = 1, \dots, K^{(l)}, \quad l = 1, \dots, L, \quad (20)$$

where:

$$\text{ReLU}(a) = \begin{cases} 0, & \text{if } a < 0, \\ a, & \text{otherwise.} \end{cases} \quad (21)$$

The final output is given by:

$$g(z_{i,t}; \theta) = \left(x_{i,t}^{(L)} \right)' \theta^{(L)}. \quad (22)$$

Here, $K^{(l)}$ denotes the number of neurons in hidden layer $l = 1, \dots, L$ and $x_{i,t,k}^{(l)}$ denotes the output of neuron k in layer l . The input layer is given by the augmented characteristic vector $x_{i,t}^{(0)} = (1, z_{i,t,1}, \dots, z_{i,t,N})'$, with $K^{(0)} = N$. For each hidden layer, the output vector is augmented with a constant term, $x_{i,t}^{(l)} = \left(1, x_{i,t,1}^{(l)}, \dots, x_{i,t,K^{(l)}}^{(l)} \right)'$. Since each neuron in layer l receives $1 + K^{(l-1)}$ inputs, including the constant term, hidden layer l contains $K^{(l)} (1 + K^{(l-1)})$ trainable parameters and the output layer adds a further $1 + K^{(L)}$ trainable parameters.

The FFNN is regularised through an L_1 penalty applied to all trainable parameters:

$$\phi(\theta; \lambda_1) = \lambda_1 \sum_{p \in \Theta} |p|, \quad (23)$$

where Θ denotes the set of all trainable network parameters, including weights and biases and $\lambda_1 \geq 0$ controls the strength of regularisation. The FFNN parameters are therefore estimated by minimising

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{T}|} \sum_{(i,t) \in \mathcal{T}} (r_{i,t+1} - g(z_{i,t}; \theta))^2 + \phi(\theta; \lambda_1). \quad (24)$$

The L_1 penalty shrinks all trainable parameters towards zero, thereby reducing overfitting risk in the high-dimensional characteristic space.

Optimisation is performed using Adam⁷ with a batch size of 10 000 and a learning rate of 0.001. Training is capped at 50 epochs and regularised through validation-based early stopping with patience equal to three epochs. To reduce sensitivity to random initialisation and stochastic optimisation, three independently initialised networks are trained for each specification and their forecasts are averaged:

$$\hat{r}_{i,t+1}^{\text{ens}} = \frac{1}{3} \sum_{m=1}^3 g(z_{i,t}; \theta_m).$$

A.5. Ensemble-Based Models

As demonstrated by Rasekhschaffe and Jones (2019) and Bali et al. (2023)), ensemble models enhance predictive performance by aggregating information from multiple learners. Even when individual components exhibit limited predictive power on their own, the combined structure captures patterns and relationships, resulting in superior forecasting. To implement this framework, a simple equal-weighted model will be created, following the methodology outlined in Bali et al. (2023).

Accordingly, two ensemble models will be created, a linear and non-linear ensemble, constructed from the individual forecasting models within their respective class:

$$\hat{r}_{i,t+1}^{En} = \frac{1}{J} \sum_{j \in \mathcal{J}} \hat{r}_{i,t+1}^{(j)}, \quad (25)$$

where $\mathcal{J}$ is the set of target models in the ensemble, J is the number of models in the framework and $\hat{r}_{i,t+1}^{(j)}$ is the predicted excess return for stock i at period $t + 1$ from model j . The underlying assumption is that although individual models may be weak learners, aggregating their forecasts will produce superior performance (Rasekhschaffe & Jones, 2019; Bali et al., 2023).

⁶See Jarrett et al. (2009) for foundational work.

⁷See Kingma and Ba (2015) for foundational work.

B. Characteristics Rank Transformation

This paper applies the methodology developed by Freyberger et al. (2020), where they described a way to normalise stock characteristics. The main goal is to estimate a conditional mean function m_t for S characteristics, where m_t is defined as:

$$m_t(C_{1,it-1}, \dots, C_{S,it-1}) = E[R_{it} | C_{1,it-1}, \dots, C_{S,it-1}]. \quad (26)$$

Next, for each characteristic s , a transformation function $F_{s,t}(\cdot)$, with its inverse $F_{s,t}^{-1}(\cdot)$ are used. The transformed characteristic $\tilde{C}_{s,it-1}$ and the corresponding transformed conditional mean function become:

$$\tilde{C}_{s,it-1} = F_{s,t}(C_{s,it-1}), \quad \tilde{m}_t(C_1, \dots, C_S) = m_t(F_{1,t}^{-1}(C_1), \dots, F_{S,t}^{-1}(C_S)). \quad (27)$$

This means that instead of directly estimating m_t , $\tilde{m}_t$ can be estimated, which is based on a rank transformation. This rank transformation ensures the focus lies in the relative position of stocks based on their characteristics, rather than their absolute values. To apply the transformation, each characteristic is ranked $C_{s,it-1}$ across all stocks in a given period, ensuring that the values lie between the unit interval $[0, 1]$. Formally, the rank transformation is described as:

$$F_{s,t}(C_{s,it-1}) = \frac{\text{rank}(C_{s,it-1})}{N_t + 1}, \quad (28)$$

where $\text{rank}(C_{s,it-1})$ is the rank of each stock's characteristic within the cross-section, with the smallest value assigned a rank of 1 and the largest assigned a rank of N_t . As a result, the α -quantile of the transformed characteristic $\tilde{C}_{s,it-1}$ corresponds to α , directly related to portfolio sorting and allowing for simple interpretation of relative positioning. Furthermore, dividing by $N_t + 1$ ensures that the characteristics lie within the unit interval.

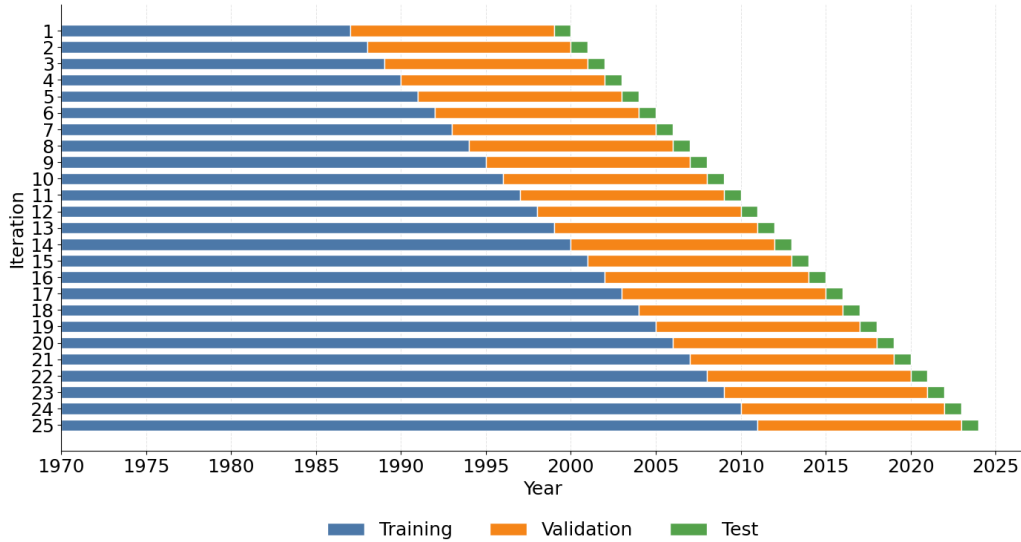
However this paper follows Gu et al. (2020) in approach, therefore the rank transformation will be modified. Ensuring the characteristics are mapped to the unit interval $[-1, 1]$ instead of $[0, 1]$. This procedure consists of two steps, first the standard rank transformation to each characteristic will be applied, in a second step a linear scaling will be introduced:

$$\tilde{C}_{s,it-1} = 2 \times \frac{\text{rank}(C_{s,it-1})}{N_t + 1} - 1. \quad (29)$$

This ensures that the values of the characteristic are mapped to the interval $[-1, 1]$, with intermediate values distributed in between.

C. Sample Split Scheme

Figure 1: Visual Representation of Sample Split Scheme



Note: This figure visualises how the models are trained every iteration, highlighting how they are re-estimated at each point in time.

D. Hyperparameter Tuning

Table 6: Hyperparameters for All Models

	OLS	OLS-FF3	PCR	IPCA	PLS	GBRT	RF	FFNN1-FFNN5
Huber loss ξ	✓	✓	-	-	-	✓	-	-
Others	-	-	K	K	K	Depth= 1 ~ 2 #Trees= {10, 50, 100, 200, 300, 500, 750, 1000} Learning rate LR $\in$ {0.01, 0.1}	Depth= 1 ~ 6 #Trees= 300 #Features in each split $\in$ {3, 5, 10, 20, 30, 50}	L1 penalty $\lambda_1 \in \{10^{-5}\}$ Learning rate LR $\in$ {0.001} Batch size = 10000 Epochs = 50 Patience = 3 Adam = Default Ensemble = 3

Note: The table lists the hyperparameters tuned for each model used in the thesis. OLS and OLS-FF3 are a special case where Huber loss is $\xi = \infty$ (see Fan et al. (2017)). All models closely follow the specifications outlined in Gu et al. (2020) and Drobetz and Otto (2021), with non-linear models hyperparameters adjusted for computational speed and sample size. A dash (-) indicates no additional model-specific hyperparameter beyond standard estimation.

E. Firm-Level Characteristics

Table 7: Firm-Level Characteristics

No.	Signal Theme	Description	Variable Name
1	Accruals	Change in current operating working capital	cowc_gr1a
2	Accruals	Operating accruals	oaccruals_at
3	Accruals	Percent operating accruals	oaccruals_ni
4	Accruals	Years 16-20 lagged returns, nonannual	seas_16_20na
5	Accruals	Total accruals	taccruals_at
6	Accruals	Percent total accruals	taccruals_ni
7	Debt Issuance	Abnormal corporate investment	capex_abn
8	Debt Issuance	Debt growth over three years	debt_gr3
9	Debt Issuance	Change in financial liabilities	fnl_gr1a
10	Debt Issuance	Change in noncurrent operating liabilities	ncol_gr1a
11	Debt Issuance	Change in net financial assets	nfna_gr1a
12	Debt Issuance	Earnings persistence	ni_ar1
13	Debt Issuance	Net operating assets	noa_at
14	Investment	Liquidity of book assets	aliq_at
15	Investment	Asset growth	at_gr1
16	Investment	Change in common equity	be_gr1a
17	Investment	CAPEX growth, one year	capx_gr1
18	Investment	CAPEX growth, two years	capx_gr2
19	Investment	CAPEX growth, three years	capx_gr3
20	Investment	Change in current operating assets	coa_gr1a
21	Investment	Change in current operating liabilities	col_gr1a
22	Investment	Hiring rate	emp_gr1
23	Investment	Inventory growth	inv_gr1
24	Investment	Inventory change	inv_gr1a
25	Investment	Change in long-term net operating assets	lnoa_gr1a
26	Investment	Mispricing factor: Management	mispricing_mgmt
27	Investment	Change in noncurrent operating assets	ncoa_gr1a
28	Investment	Change in net noncurrent operating assets	nncoa_gr1a
29	Investment	Change in net operating assets	noa_gr1a
30	Investment	Change in PPE and inventory	ppeinv_gr1a
31	Investment	Long-term reversal	ret_60_12
32	Investment	Sales growth, one year	sale_gr1
33	Investment	Sales growth, three years	sale_gr3
34	Investment	Sales growth, one quarter	saleq_gr1
35	Investment	Years 2-5 lagged returns, nonannual	seas_2_5na
36	Low Leverage	Firm age	age
37	Low Leverage	Liquidity of market assets	aliq_mat
38	Low Leverage	Book leverage	at_be
39	Low Leverage	The high-low bid-ask spread	bidaskhl_21d
40	Low Leverage	Cash-to-assets	cash_at
41	Low Leverage	Net debt-to-price	netdebt_me
42	Low Leverage	Earnings volatility	ni_ivol
43	Low Leverage	R&D-to-sales	rd_sale
44	Low Leverage	R&D capital-to-book assets	rd5_at
45	Low Leverage	Asset tangibility	tangibility
46	Low Leverage	Altman Z-score	z_score
47	Low Risk	Market beta	beta_60m
48	Low Risk	Dimson beta	beta_dimson_21d
49	Low Risk	Frazzini-Pedersen market beta	betabab_1260d
50	Low Risk	Downside beta	betadown_252d
51	Low Risk	Earnings variability	earnings_variability
52	Low Risk	Idiosyncratic volatility from the CAPM, 21 days	ivol_capm_21d
53	Low Risk	Idiosyncratic volatility from the CAPM, 252 days	ivol_capm_252d
54	Low Risk	Idiosyncratic volatility from the Fama-French three-factor model	ivol_ff3_21d
55	Low Risk	Idiosyncratic volatility from the q-factor model	ivol_hxz4_21d
56	Low Risk	Cash-flow volatility	ocfq_saleq_std
57	Low Risk	Maximum daily return	rmax1_21d
58	Low Risk	Highest five days of return	rmax5_21d
59	Low Risk	Return volatility	rvol_21d
60	Low Risk	Years 6-10 lagged returns, nonannual	seas_6_10na
61	Low Risk	Share turnover	turnover_126d
62	Low Risk	Number of zero trades with turnover as tiebreaker, one month	zero_trades_21d
63	Low Risk	Number of zero trades with turnover as tiebreaker, six months	zero_trades_126d
64	Low Risk	Number of zero trades with turnover as tiebreaker, twelve months	zero_trades_252d
65	Momentum	Current price to high price over last year	prc_highprc_252d
66	Momentum	Price momentum, $t - 3$ to $t - 1$	ret_3_1
67	Momentum	Price momentum, $t - 6$ to $t - 1$	ret_6_1
68	Momentum	Price momentum, $t - 9$ to $t - 1$	ret_9_1
69	Momentum	Price momentum, $t - 12$ to $t - 1$	ret_12_1
70	Momentum	Residual momentum, $t - 6$ to $t - 1$	resff3_6_1
71	Momentum	Residual momentum, $t - 12$ to $t - 1$	resff3_12_1
72	Momentum	Year 1-lagged return, nonannual	seas_1_1na

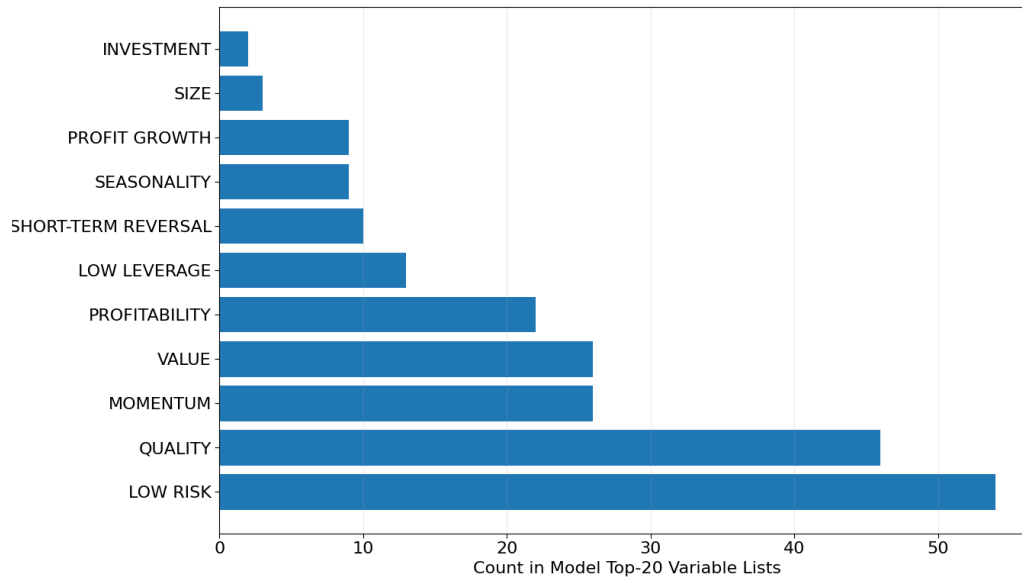
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No.	Signal Theme	Description	Variable Name
73	Profit Growth	Change sales minus change inventory	dsale_dinv
74	Profit Growth	Change sales minus change receivables	dsale_drec
75	Profit Growth	Change sales minus change SG&A	dsale_dsga
76	Profit Growth	Change in quarterly return on assets	niq_at_chg1
77	Profit Growth	Change in quarterly return on equity	niq_be_chg1
78	Profit Growth	Standardised earnings surprise	niq_su
79	Profit Growth	Change in operating cash flow to assets	ocf_at_chg1
80	Profit Growth	Price momentum, $t - 12$ to $t - 7$	ret_12_7
81	Profit Growth	Labour-force efficiency	sale_emp_gr1
82	Profit Growth	Standardised revenue surprise	saleq_su
83	Profit Growth	Year 1-lagged return, annual	seas_1_1an
84	Profit Growth	Tax expense surprise	tax_gr1a
85	Profitability	Coefficient of variation for dollar trading volume	dolvol_var_126d
86	Profitability	Return on net operating assets	ebit_bev
87	Profitability	Profit margin	ebit_sale
88	Profitability	Piotroski F-score	f_score
89	Profitability	Return on equity	ni_be
90	Profitability	Quarterly return on equity	niq_be
91	Profitability	Ohlson O-score	o_score
92	Profitability	Operating cash flow to assets	ocf_at
93	Profitability	Operating profits-to-book equity	ope_be
94	Profitability	Operating profits-to-lagged book equity	ope_bell
95	Profitability	Coefficient of variation for share turnover	turnover_var_126d
96	Quality	Capital turnover	at_turnover
97	Quality	Cash-based operating profits-to-book assets	cop_at
98	Quality	Cash-based operating profits-to-lagged book assets	cop_atl1
99	Quality	Change gross margin minus change sales	dgp_dsale
100	Quality	Gross profits-to-assets	gp_at
101	Quality	Gross profits-to-lagged assets	gp_atl1
102	Quality	Mispricing factor: Performance	mispricing_perf
103	Quality	Number of consecutive quarters with earnings increases	ni_inc8q
104	Quality	Quarterly return on assets	niq_at
105	Quality	Operating profits-to-book assets	op_at
106	Quality	Operating profits-to-lagged book assets	op_atl1
107	Quality	Operating leverage	opex_at
108	Quality	Quality minus Junk: Composite	qmj
109	Quality	Quality minus Junk: Growth	qmj_growth
110	Quality	Quality minus Junk: Profitability	qmj_prof
111	Quality	Quality minus Junk: Safety	qmj_safety
112	Quality	Asset turnover	sale_bev
113	Seasonality	Market correlation	corr_1260d
114	Seasonality	Coskewness	coskew_21d
115	Seasonality	Net debt issuance	dbnetis_at
116	Seasonality	Kaplan-Zingales index	kz_index
117	Seasonality	Change in long-term investments	lti_gr1a
118	Seasonality	Taxable income-to-book income	pi_nix
119	Seasonality	Years 2-5 lagged returns, annual	seas_2_5an
120	Seasonality	Years 6-10 lagged returns, annual	seas_6_10an
121	Seasonality	Years 11-15 lagged returns, annual	seas_11_15an
122	Seasonality	Years 11-15 lagged returns, nonannual	seas_11_15na
123	Seasonality	Years 16-20 lagged returns, annual	seas_16_20an
124	Seasonality	Change in short-term investments	sti_gr1a
125	Size	Amihud measure	ami_126d
126	Size	Dollar trading volume	dolvol_126d
127	Size	Market equity	market_equity
128	Size	Price per share	prc
129	Size	R&D-to-market	rd_me
130	Short-term Reversal	Idiosyncratic skewness from the CAPM	iskew_capm_21d
131	Short-term Reversal	Idiosyncratic skewness from the Fama-French three-factor model	iskew_ff3_21d
132	Short-term Reversal	Idiosyncratic skewness from the q-factor model	iskew_hxz4_21d
133	Short-term Reversal	Short-term reversal	ret_1_0
134	Short-term Reversal	Highest five days of return scaled by volatility	rmax5_rvol_21d
135	Short-term Reversal	Total skewness	rskew_21d
136	Value	Assets-to-market	at_me
137	Value	Book-to-market equity	be_me
138	Value	Book-to-market enterprise value	bev_me
139	Value	Net stock issues	chcsho_12m
140	Value	Debt-to-market	debt_me
141	Value	Dividend yield	div12m_me
142	Value	EBITDA-to-market enterprise value	ebitda_me
143	Value	Equity duration	eq_dur
144	Value	Net equity issuance	eqnetis_at
145	Value	Equity net payout	eqnpo_12m
146	Value	Net payout yield	eqnpo_me
147	Value	Payout yield	eqpo_me
148	Value	Free cash flow-to-price	fcf_me
149	Value	Intrinsic value-to-market	ival_me
150	Value	Net total issuance	netis_at
151	Value	Earnings-to-price	ni_me
152	Value	Operating cash flow-to-market	ocf_me
153	Value	Sales-to-market	sale_me

Note: This table reports the full set of firm-level characteristics used in the empirical analysis, grouped by their corresponding signal themes. The variable names follow the Global Factor Data conventions. For detailed definitions, construction methods and source references for each characteristic, see the documentation available on the Global Factor Data website.

F. Model Performance

Figure 2: Signal Theme Importance

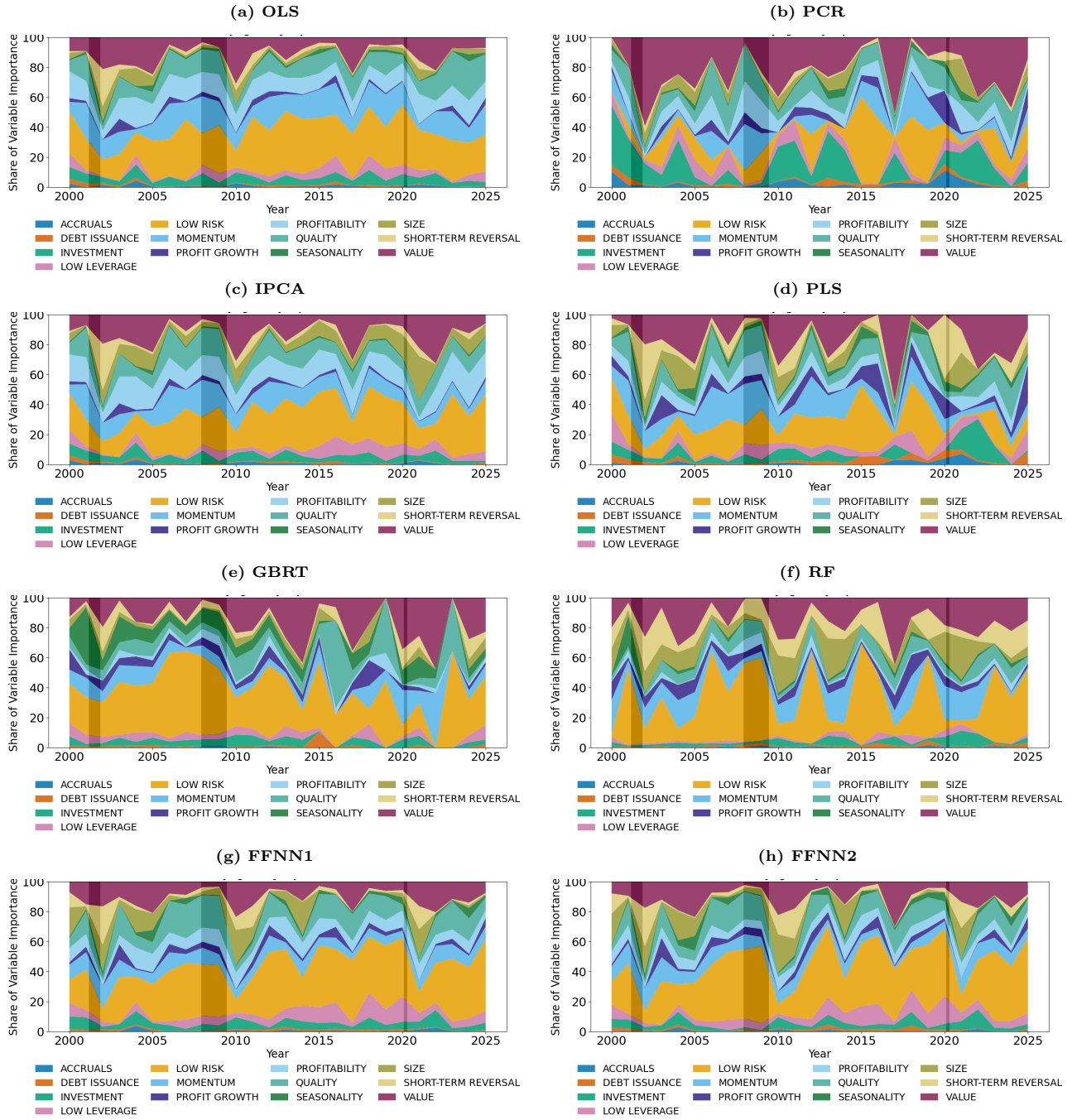


Note: This figure first ranks the 20 most important signals for each model according to the variable-importance measure defined in Equation 5. Each signal is then assigned to its corresponding signal theme to summarise the thematic composition of predictive information.

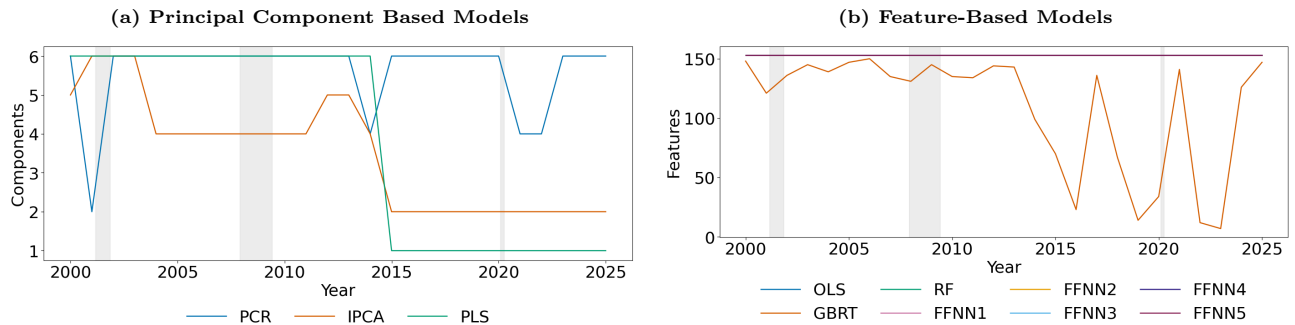
Table 8: Most Predictive Signal Themes

Signal Theme	Economic Intuition
Low Risk	Measures in this category often proxy for illiquidity, volatility, investor participation and trading frictions. Their relevance suggests that expected returns partly reflect compensation for holding securities that are more difficult or costly to trade.
Quality	Revenue- and operating based signals identify firms with stronger underlying fundamentals relative to their market valuation. Their predictive content is consistent with the idea that persistent improvements in firm quality are incorporated into prices only gradually.
Momentum	Signals that capture persistence in past performance. This pattern is consistent with gradual information diffusion, investor underreaction and delayed price adjustment following firm-specific news.
Value	Valuation ratios compare market prices with fundamentals. Their predictive power may reflect either risk compensation for firms exposed to distress and financial constraints, or the gradual correction of prices that have moved away from fundamentals.
Profitability	Profitability related characteristics reflect firms' ability to generate earnings or operating profits from their asset base, equity capital or market value. Their importance suggests that profitable firms contain persistent cash-flow information that is not fully reflected in contemporaneous prices.

Note: This table summarises the main predictive signal themes identified across models and reports the corresponding economic intuition for their return forecasting relevance.

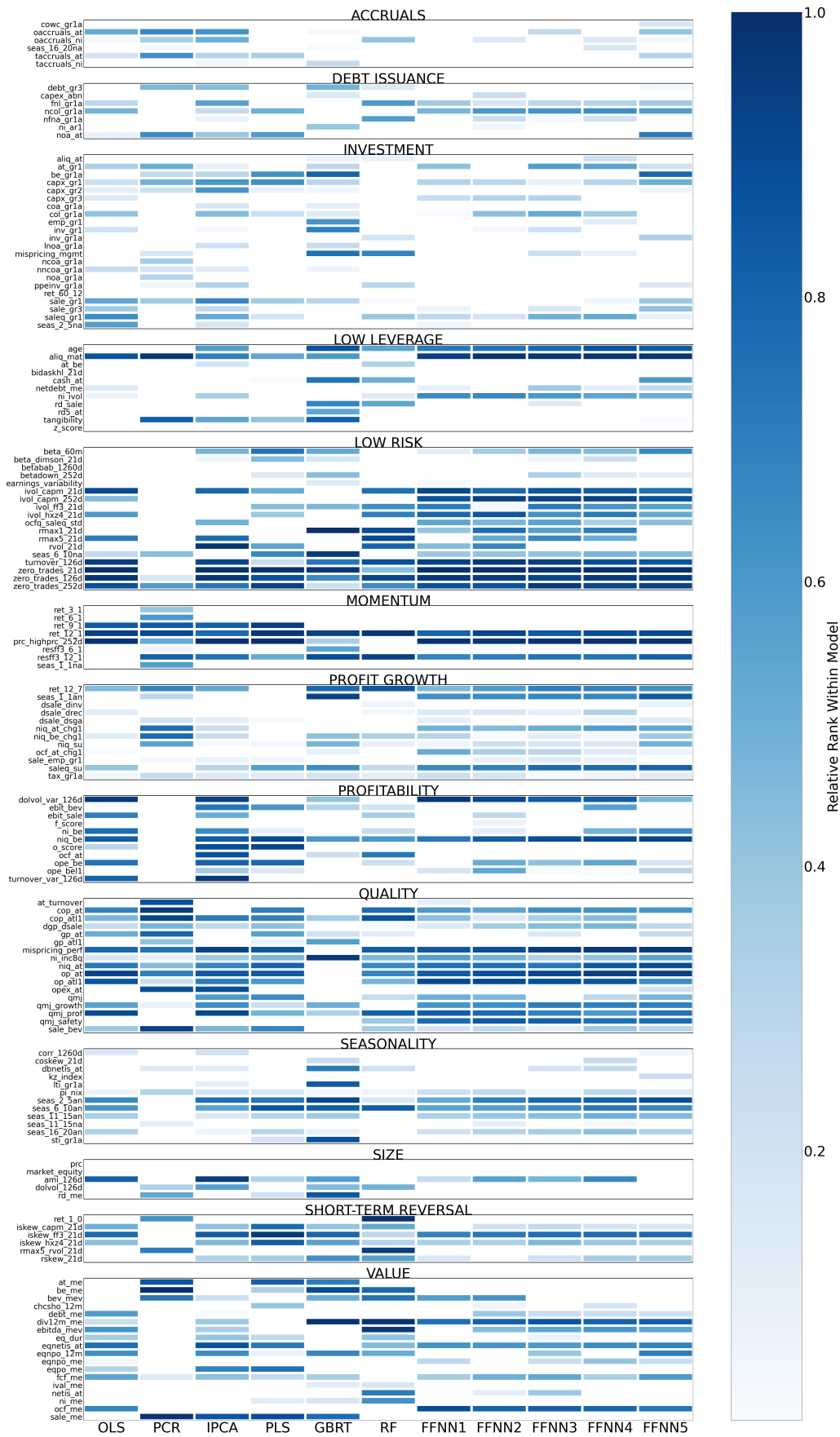
Figure 3: Time-Varying Variable Importance

Note: This figure reports the time-varying variable importance of the most influential firm-level characteristics across models in (%). Shaded areas indicate U.S. recessions as defined by the Federal Reserve Bank of St. Louis. Due to limited space, only the two best performing FFNN architectures are displayed.

Figure 4: Model Complexity

Note: This figure illustrates the time-varying model complexity estimated by each model, measured by the number of components or features used at each re-estimation point. Shaded areas indicate U.S. recessions as defined by the Federal Reserve Bank of St. Louis.

Figure 5: Firm-Level Characteristic Importance



Note: This figure shows the relative importance of the firm-level characteristics per model grouped by theme and enclosed by black lines. Characteristics with zero or negative average explanatory power as calculated by Equation 5 are excluded in the ranking to preserve visual clarity and interpretability.

G. Portfolio Performance

Table 9: Portfolio Performance: Macroeconomic Variables

Variable	Definition
Dividend-price ratio (dp)	Difference between the log of dividends and the log of prices. Dividends are calculated as the 12-month moving sums of dividends paid on the S&P 500 index.
Earnings-price ratio (ep)	Difference between the log of earnings and the log of prices. Earnings are calculated as the 12-month moving sums of earnings on the S&P 500 index.
Book-to-market ratio (bm)	Ratio of book value to market value for the Dow Jones Industrial Average.
Net equity expansion (ntis)	Measure of corporate issuing activity, defined as the ratio of 12-month moving sums of net issues by NYSE-listed stocks to the total end-of-year market capitalisation of NYSE stocks.
Term spread (tms)	Difference between the long-term yield on government bonds and the Treasury bill rate.
Default spread (dfy)	Difference between BAA- and AAA-rated corporate bond yields.
Stock variance (svar)	Calculated as the sum of squared daily returns on the S&P 500.

Note: This table reports the macroeconomic variables obtained from Welch and Goyal (2008) and used in the macro regressions.

Table 10: Portfolio Performance: Risk-Adjusted Performance and Macro Exposures of Long-Short Portfolios

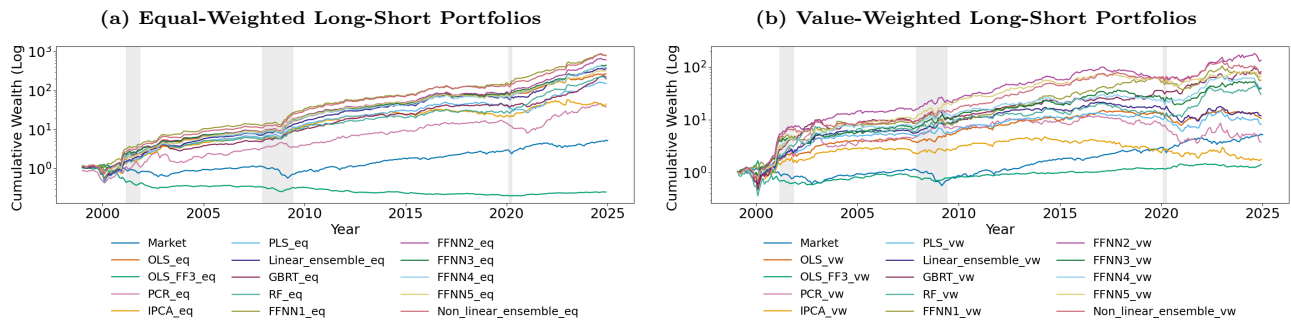
Panel A: Linear Models								
	OLS	OLS-FF3	PCR	IPCA	PLS	Linear-En		
Drawdown and Turnover: Equal-Weighted								
Max DD (%)	52.19	84.45	52.98	56.63	60.34	57.13		
Max 1M loss (%)	34.93	19.27	34.79	37.69	39.23	35.07		
Turnover (%)	275.57	28.07	102.86	236.47	232.82	247.81		
Drawdown and Turnover: Value-Weighted								
Max DD (%)	47.26	64.93	70.33	64.44	55.37	58.70		
Max 1M loss (%)	29.09	21.03	23.47	34.45	31.57	28.66		
Turnover (%)	318.12	9.40	101.23	268.75	260.12	283.07		
Risk-Adjusted Performance: Equal-Weighted								
FF5 α	1.68*** (4.32)	-0.22 (-1.36)	1.18*** (4.59)	1.28*** (3.14)	1.52*** (4.25)	1.78*** (5.43)		
FF5 R^2	0.10	0.46	0.70	0.06	0.29	0.24		
CAPM α	1.88*** (4.89)	-0.56** (-2.39)	1.98*** (4.49)	1.43*** (3.79)	1.89*** (4.57)	2.15*** (5.66)		
CAPM R^2	0.02	0.13	0.27	0.00	0.01	0.01		
Risk-Adjusted Performance: Value-Weighted								
FF5 α	0.50* (1.78)	0.29* (1.91)	0.54** (2.08)	0.10 (0.32)	0.67** (2.24)	0.58** (2.11)		
FF5 R^2	0.17	0.53	0.71	0.11	0.23	0.27		
CAPM α	0.85** (2.47)	0.00 (0.02)	1.43*** (3.46)	0.45 (1.41)	1.08*** (3.07)	1.14*** (3.18)		
CAPM R^2	0.01	0.12	0.38	0.00	0.04	0.03		
Macro Regressions: Equal-Weighted								
α	0.73** (2.05)	-0.04 (-0.23)	0.02 (0.06)	0.69* (1.83)	0.66* (1.84)	0.67** (2.22)		
dp	-0.14** (-2.03)	0.02 (0.73)	-0.03 (-0.45)	-0.13* (-1.76)	-0.13* (-1.94)	-0.13** (-2.21)		
ep	-0.01 (-1.18)	-0.01** (-2.35)	0.02* (1.65)	-0.02 (-1.62)	-0.00 (-0.41)	-0.01 (-0.71)		
bm	0.13 (1.03)	0.01 (0.08)	-0.01 (-0.09)	0.15 (1.12)	0.13 (1.03)	0.12 (1.08)		
ntis	-0.22 (-0.52)	0.07 (0.42)	-0.00 (-0.01)	-0.11 (-0.25)	-0.14 (-0.31)	-0.21 (-0.54)		
tms	0.09 (0.65)	-0.04 (-0.62)	-0.15 (-1.27)	0.05 (0.36)	-0.12 (-1.10)	-0.03 (-0.26)		
dfy	5.75*** (4.49)	-0.04 (-0.05)	0.63 (0.32)	4.65*** (3.92)	5.27*** (3.88)	4.83*** (3.71)		
svar	-1.08 (-1.56)	-0.90* (-1.80)	0.53 (0.58)	-0.58 (-0.84)	-0.83 (-1.20)	-0.59 (-0.87)		
Macro R^2	0.08	0.05	0.02	0.06	0.06	0.06		
Macro Regressions: Value-Weighted								
α	0.24 (0.71)	0.23 (1.20)	-0.48 (-1.25)	0.32 (0.92)	0.14 (0.37)	0.20 (0.57)		
dp	-0.07 (-1.08)	-0.03 (-0.92)	0.05 (0.74)	-0.08 (-1.22)	-0.06 (-0.76)	-0.06 (-1.00)		
ep	0.01 (1.15)	-0.01* (-1.92)	0.04** (2.34)	0.01 (1.06)	0.02 (1.46)	0.02 (1.36)		
bm	-0.03 (-0.30)	0.10 (1.36)	-0.26 (-1.54)	0.04 (0.35)	-0.05 (-0.35)	-0.05 (-0.42)		
ntis	0.22 (0.61)	-0.02 (-0.08)	0.59 (1.41)	0.10 (0.26)	0.37 (0.85)	0.29 (0.71)		
tms	0.04 (0.36)	0.07 (1.09)	-0.11 (-0.76)	0.14 (1.10)	-0.14 (-1.03)	-0.04 (-0.41)		
dfy	5.64*** (4.26)	0.42 (0.55)	2.97 (1.28)	3.53*** (3.17)	3.49** (2.15)	5.25*** (4.11)		
svar	-1.67*** (-2.68)	-1.01* (-1.96)	1.52* (1.89)	0.03 (0.05)	0.69 (0.80)	-0.10 (-0.14)		
Macro R^2	0.06	0.05	0.05	0.03	0.04	0.04		
Panel B: Non-Linear Models								
	GBRT	RF	FFNN1	FFNN2	FFNN3	FFNN4	FFNN5	Non-Linear-En
Drawdown and Turnover: Equal-Weighted								
Max DD (%)	56.53	58.70	15.46	33.61	31.24	37.10	37.02	40.71
Max 1M loss (%)	31.89	36.59	14.38	21.90	22.33	23.03	23.23	25.45
Turnover (%)	277.01	230.16	254.47	254.22	256.27	259.15	261.23	258.16
Drawdown and Turnover: Value-Weighted								
Max DD (%)	60.04	65.04	47.15	45.16	47.14	42.02	41.80	45.93

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	GBRT	RF	FFNN1	FFNN2	FFNN3	FFNN4	FFNN5	Non-Linear-En
Max 1M loss (%)	32.40	34.83	17.20	16.75	20.89	17.96	23.30	24.26
Turnover (%)	310.05	291.10	291.35	297.12	297.37	296.69	298.11	301.74
Risk-Adjusted Performance: Equal-Weighted								
FF5 α	1.56*** (5.15)	1.66*** (4.85)	2.14*** (5.41)	2.04*** (4.83)	1.98*** (4.84)	1.89*** (4.66)	1.82*** (4.72)	2.11*** (5.05)
FF5 R^2	0.36	0.28	0.04	0.05	0.04	0.05	0.05	0.07
CAPM α	1.94*** (4.99)	2.02*** (4.80)	2.20*** (6.05)	2.13*** (5.55)	2.03*** (5.58)	1.98*** (5.46)	1.94*** (5.59)	2.27*** (5.67)
CAPM R^2	0.01	0.01	0.03	0.02	0.02	0.02	0.01	0.01
Risk-Adjusted Performance: Value-Weighted								
FF5 α	1.23*** (3.86)	0.75** (2.30)	1.33*** (2.76)	1.58*** (3.29)	1.19*** (2.77)	1.10** (2.46)	1.27*** (3.28)	1.24*** (2.77)
FF5 R^2	0.48	0.50	0.07	0.10	0.09	0.10	0.12	0.21
CAPM α	1.96*** (4.04)	1.67*** (2.90)	1.57*** (3.03)	1.92*** (3.60)	1.46*** (3.10)	1.47*** (3.11)	1.64*** (3.57)	1.77*** (3.28)
CAPM R^2	0.08	0.05	0.00	0.01	0.00	0.00	0.01	0.01
Macro Regressions: Equal-Weighted								
α	0.71** (2.43)	0.95*** (3.18)	0.67* (1.93)	0.78** (1.99)	0.79** (2.15)	0.68* (1.93)	0.64* (1.92)	0.78** (2.03)
dp	-0.15*** (-2.73)	-0.17*** (-2.89)	-0.12* (-1.82)	-0.15** (-1.99)	-0.14** (-1.87)	-0.13* (-1.88)	-0.12* (-1.87)	-0.15** (-2.01)
ep	0.00 (0.14)	-0.02** (-2.49)	-0.02 (-1.58)	-0.01 (-1.12)	-0.02* (-1.88)	-0.01 (-1.21)	-0.01 (-1.36)	-0.01 (-1.31)
bm	0.21** (2.01)	0.19* (1.81)	0.09 (0.84)	0.14 (1.06)	0.16 (1.32)	0.11 (0.93)	0.12 (1.07)	0.15 (1.24)
ntis	-0.61* (-1.65)	-0.59 (-1.59)	-0.21 (-0.56)	-0.41 (-0.96)	-0.36 (-0.91)	-0.27 (-0.69)	-0.22 (-0.59)	-0.37 (-0.87)
tms	0.12 (1.24)	0.09 (0.68)	0.21 (1.52)	0.20 (1.33)	0.18 (1.28)	0.18 (1.34)	0.19 (1.39)	0.20 (1.38)
dfy	4.27*** (3.43)	3.13*** (2.69)	5.12*** (4.25)	5.51*** (4.23)	5.01*** (4.01)	5.73*** (4.41)	5.54*** (4.75)	5.28*** (4.20)
svar	-0.14 (-0.22)	0.01 (0.02)	-0.49 (-0.69)	-0.54 (-0.78)	-0.57 (-0.88)	-0.82 (-1.37)	-0.94 (-1.56)	-0.29 (-0.40)
Macro R^2	0.07	0.07	0.10	0.09	0.09	0.10	0.10	0.09
Macro Regressions: Value-Weighted								
α	0.15 (0.39)	0.50 (1.19)	0.19 (0.40)	0.34 (0.63)	0.19 (0.38)	0.03 (0.06)	0.26 (0.68)	0.23 (0.45)
dp	-0.04 (-0.57)	-0.10 (-1.22)	-0.06 (-0.69)	-0.10 (-0.97)	-0.06 (-0.62)	-0.03 (-0.34)	-0.06 (-0.86)	-0.06 (-0.66)
ep	0.02 (1.06)	0.00 (0.18)	0.03* (1.92)	0.03* (1.81)	0.02 (1.27)	0.03** (2.11)	0.01 (0.63)	0.02 (1.23)
bm	-0.15 (-0.94)	-0.08 (-0.52)	-0.09 (-0.59)	-0.08 (-0.49)	-0.05 (-0.35)	-0.23 (-1.40)	-0.06 (-0.47)	-0.08 (-0.46)
ntis	0.16 (0.38)	0.03 (0.06)	0.02 (0.04)	0.05 (0.09)	0.12 (0.24)	0.27 (0.52)	0.04 (0.09)	0.13 (0.24)
tms	-0.17 (-1.43)	-0.26 (-1.62)	0.11 (0.66)	-0.12 (-0.67)	-0.14 (-0.70)	-0.19 (-1.08)	-0.05 (-0.28)	-0.20 (-1.20)
dfy	2.68 (1.36)	4.85*** (2.75)	1.94 (3.04)	5.68*** (3.04)	4.60*** (2.60)	5.76*** (3.05)	5.90*** (4.58)	4.35** (2.44)
svar	-0.34 (-0.35)	-1.05 (-1.12)	-0.51 (-0.59)	-1.00 (-0.93)	-1.35 (-1.36)	-1.29 (-1.18)	-2.07** (-2.18)	-0.98 (-0.84)
Macro R^2	0.04	0.05	0.03	0.05	0.03	0.05	0.06	0.03

Note: This table reports drawdown, turnover, risk-adjusted performance and macroeconomic exposures for long-short portfolios. Maximum drawdown, maximum one-month loss and turnover are reported separately for equal-weighted and value-weighted portfolio construction. Risk-adjusted performance is evaluated using CAPM and FF5 alphas, with corresponding t-statistics reported in parentheses and R^2 values reported separately. Macro regressions report exposures to the macroeconomic variables. Standard errors are computed using Newey and West (1987) with four lags, following Drobetz and Otto (2021). Significance at the 1%, 5%, and 10% levels is denoted by ***, ** and *, respectively.

Figure 6: Portfolio Performance: Cumulative Net Returns of Long-Short Portfolios



Note: Cumulative returns of portfolio out-of-sample performance from 1999 to 2024 with shaded areas indicating U.S. recessions as defined by the Federal Reserve Bank of St. Louis. Market excess returns are taken from Kenneth French's website.

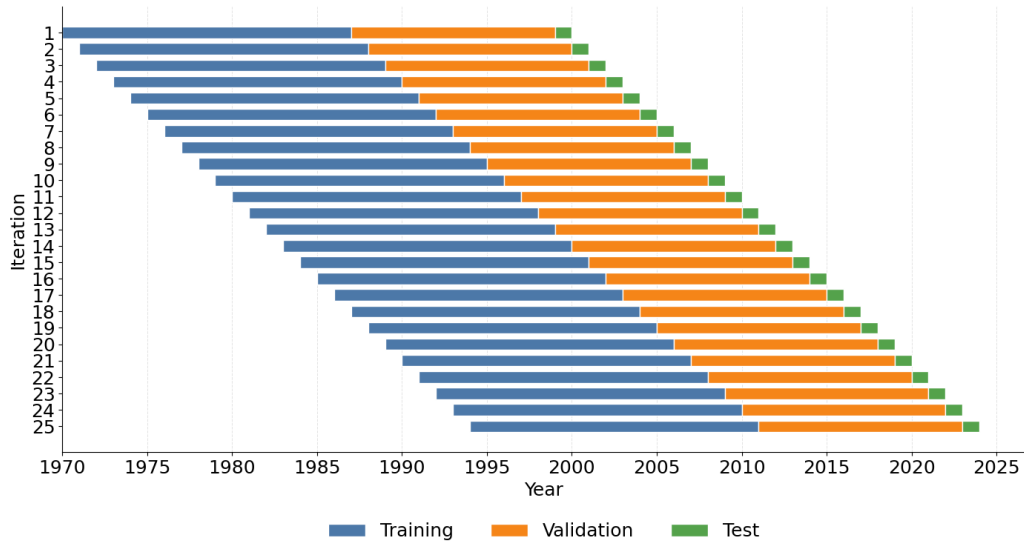
H. Robustness Checks

Table 11: Robustness Checks:n Robustness Settings

Robustness Settings	Description	Intuition
Winsorisation	Extreme return observations are winsorised cross-sectionally at the 5% level within each month t , prior to applying the rank transformation.	Winsorisation is used as a robustness check to assess whether the results are driven by extreme observations. While rank normalisation standardises characteristics by their cross-sectional ordering, it does not remove the influence of tail observations on the underlying return distribution or on relationships formed prior to ranking. Cross-sectional winsorisation therefore limits the impact of unusually large return realisations while preserving the relative structure of the data.
Alternative characteristics database	The Jensen et al. (2023) characteristics library is replaced with the original Gu et al. (2020) dataset obtained from Dacheng Xiu's website.	The alternative dataset serves as a robustness check for data-source dependence. It assesses whether the results are driven by the specific construction choices in the Jensen et al. (2023) characteristics library or instead reflect return-predictive information that remains stable across alternative characteristic definitions and signal construction methodologies.
Market capitalisation segmentation	Each model is re-estimated separately for the largest and smallest 1000 stocks by market capitalisation.	Market capitalisation segmentation evaluates whether the findings are robust across firm-size groups rather than driven by a specific segment.
Alternative industry classification	The SIC two-digit specification is substituted for the Fama and French (1997) 48-industry classification.	Alternative industry classification is used to assess whether the results are sensitive to the industry specification embedded in the empirical design. This robustness check tests whether the findings reflect genuine predictive structure rather than dependence on a particular way of grouping firms across industries.
Alternative sample split scheme	The original sample-splitting procedure is replaced by a fixed-rolling sample split. Under the fixed-rolling scheme, the initial train, validation and test window lengths are preserved, while the training and validation windows move forward through time, maintaining a constant sample length and progressively incorporating more recent observations. See Figure 7 for a visualisation of the alternative split.	The alternative sample split scheme tests whether the results are sensitive to the specific train-validation-test design. Forecasting performance may depend on the initial training period, validation window and market regimes included in the sample. This check evaluates whether the findings reflect genuine predictive structure rather than partition-specific overfitting or timing effects.

Note: This table summarises the robustness checks used to assess whether the main results are sensitive to outliers, data source selection, market capitalisation segmentation, industry classification or sample split scheme.

Figure 7: Robustness Checks: Visual Representation of Alternative Sample Split Scheme



Note: This figure visualises how the models are trained every iteration, highlighting how they are re-estimated at each point in time for the alternative sample split.

Table 12: Robustness Checks: Adjusted DM Test Comparison of Out-of-Sample Predictive Performance

	OLS-FF3	PCR	IPCA	PLS	Linear-En	GBRT	RF	FFNN1	FFNN2	FFNN3	FFNN4	FFNN5	Non-Linear-En
Panel A: Winsorisation													
OLS	-0.494	-0.663	-0.997	1.791	2.080	0.783	3.911*	0.715	0.779	0.681	0.623	1.545	4.510*
OLS-FF3		0.153	0.124	0.706	1.313	0.715	1.119	0.638	0.671	0.628	0.590	0.801	1.145
PCR			-0.134	1.260	2.445	1.112	1.884	0.739	0.779	0.734	0.730	1.127	1.725
IPCA				1.780	3.221*	1.396	2.310	1.028	1.051	0.996	1.011	1.413	2.306
PLS					1.401	0.225	2.455	-0.115	-0.041	-0.198	-0.452	0.304	2.218
Linear-En						-0.604	0.076	-0.878	-0.828	-0.965	-1.251	-0.918	0.280
GBRT							0.801	-0.256	-0.201	-0.299	-0.437	-0.042	1.036
RF								-1.251	-1.196	-1.452	-2.093	-2.037	0.402
FFNN1									0.204	-0.170	-0.460	0.333	2.891*
FFNN2										-0.539	-0.699	0.282	2.946*

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	OLS-FF3	PCR	IPCA	PLS	Linear-En	GBRT	RF	FFNN1	FFNN2	FFNN3	FFNN4	FFNN5	Non-Linear-En
FFNN3											-0.442	0.497	3.665*
FFNN4												0.977	5.854*
FFNN5													2.213
Panel B: Alternative Characteristic Database													
OLS	0.557	-1.580	-1.252	0.838	1.246	-1.785	1.427	-1.181	0.131	0.741	0.139	0.583	0.827
OLS-FF3		-1.143	-1.007	-0.522	-0.274	-1.133	-0.373	-0.719	-0.528	-0.427	-0.513	-0.460	-0.439
PCR			1.884	1.666	2.980*	-0.071	1.707	0.917	1.283	1.432	1.311	1.549	1.604
IPCA				1.336	2.598	-0.439	1.407	0.648	1.025	1.184	1.044	1.254	1.313
PLS					1.114	-1.965	1.179	-1.385	-0.094	0.528	-0.086	0.398	0.614
Linear-En						-2.381	-0.464	-1.412	-0.868	-0.575	-0.836	-0.723	-0.682
GBRT							2.227	1.187	1.696	1.889	1.816	2.054	2.297
RF								-2.773	-1.418	-0.480	-1.381	-0.668	-0.828
FFNN1									2.581	3.104*	2.097	1.464	2.526
FFNN2										1.485	0.017	0.485	0.972
FFNN3											-1.361	-0.194	-0.123
FFNN4												0.533	1.241
FFNN5													0.145
Panel C: Top Market-Capitalisation Segment													
OLS	-0.201	-0.764	-1.614	1.469	0.611	-1.427	1.283	-1.343	-0.462	-0.146	-0.171	-0.031	0.405
OLS-FF3		-0.399	-1.011	0.577	0.622	-0.134	0.637	-0.516	-0.004	0.174	0.146	0.230	0.414
PCR			-2.849	1.147	1.792	0.327	1.273	0.023	0.530	0.730	0.679	0.692	0.987
IPCA				1.994	3.246*	1.101	2.233	0.935	1.402	1.629	1.584	1.506	1.918
PLS					-0.165	-2.217	0.117	-2.659	-1.689	-1.258	-1.135	-0.905	-0.794
Linear-En						-1.133	0.252	-2.323	-1.133	-0.860	-0.824	-0.573	-0.331
GBRT							2.139	-0.514	0.267	0.545	0.459	0.549	1.136
RF								-3.383	-2.108	-1.710	-1.513	-1.153	-1.318
FFNN1									2.710	3.989*	3.463*	2.567	4.877*
FFNN2										1.327	0.987	1.231	2.836
FFNN3											-0.165	0.283	1.922
FFNN4												0.475	1.491
FFNN5													0.804
Panel D: Bottom Market-Capitalisation Segment													
OLS	0.450	-0.343	0.567	2.091	2.909*	0.760	1.670	-2.098	-0.696	2.305	3.883*	4.479*	6.010*
OLS-FF3		-0.646	-0.080	0.177	1.073	0.092	-0.046	-1.104	-0.576	0.156	0.442	0.575	0.707
PCR			1.748	1.377	2.865	1.889	0.966	-0.449	0.127	0.906	1.208	1.680	1.717
IPCA				0.619	2.739	0.634	0.085	-1.398	-0.745	0.369	0.787	1.276	1.476
PLS					2.692	-0.132	-1.116	-2.556	-1.873	0.011	0.791	2.136	2.381
Linear-En						-1.671	-2.898	-2.914	-2.509	-1.167	-0.745	-0.919	-0.329
GBRT							-0.255	-1.540	-0.894	0.089	0.462	0.753	1.008
RF								-2.144	-1.321	0.496	1.273	2.924*	2.796
FFNN1									2.555	5.739*	6.255*	3.693*	5.566*
FFNN2										4.040*	5.870*	3.353*	5.592*
FFNN3											1.675	1.148	2.535
FFNN4												0.332	1.466
FFNN5													0.956
Panel E: Alternative Industry Classification													
OLS	-0.290	-0.436	-0.774	1.683	2.126	0.273	3.777*	0.539	1.782	1.127	1.335	2.856	5.289*
OLS-FF3		0.059	-0.021	0.499	1.140	0.346	0.899	0.419	0.739	0.530	0.540	0.784	1.036
PCR			-0.356	0.955	2.294	0.582	1.597	0.510	0.993	0.707	0.770	1.270	1.638
IPCA				1.449	3.120*	0.871	2.015	0.777	1.393	1.008	1.110	1.676	2.212
PLS					1.576	-0.444	2.514	-0.175	0.788	0.187	0.244	1.428	2.740
Linear-En						-1.321	-0.108	-1.008	-0.348	-0.792	-0.835	-0.491	0.380
GBRT							1.695	0.150	0.913	0.446	0.465	1.020	2.206
RF								-1.283	-0.362	-1.127	-1.201	-0.748	0.891
FFNN1									2.730	0.835	0.690	1.022	3.513*
FFNN2										-1.563	-1.133	-0.005	1.793
FFNN3											0.069	0.875	3.980*
FFNN4												0.910	3.797*
FFNN5													1.723
Panel F: Alternative Sample Split													
OLS	-0.772	-0.524	-0.874	2.043	2.055	0.934	4.122*	0.355	1.707	1.874	2.410	1.773	5.675*
OLS-FF3		0.433	0.370	1.031	1.672	1.056	1.465	0.833	1.090	1.159	1.210	1.213	1.620
PCR			-0.339	1.121	2.238	1.213	1.770	0.533	1.005	1.064	1.156	1.182	1.929
IPCA				1.645	3.093*	1.615	2.260	0.811	1.440	1.475	1.610	1.508	2.608
PLS					1.315	0.263	2.662	-0.526	0.446	0.621	0.868	0.686	3.268*
Linear-En						-0.630	0.237	-1.157	-0.520	-0.407	-0.357	-0.477	0.798
GBRT							0.922	-0.570	0.067	0.167	0.265	0.156	1.476
RF								-1.873	-0.978	-0.848	-0.885	-1.289	1.190
FFNN1									2.269	2.275	2.654	1.077	4.561*
FFNN2										0.357	0.594	0.137	3.190*
FFNN3											0.243	-0.017	3.002*
FFNN4												-0.153	4.059*
FFNN5													2.119

Note: Adjusted DM test statistic comparing out-of-sample stock-level prediction performance under multiple robustness settings. Positive values indicate the column model outperforms the row model. Bold font indicates conventional significance at the one-sided 5% level while an asterisk (*) indicates significance under Bonferroni adjustment, applying four lags in computing the Newey-West standard errors, following Drobetz and Otto (2021).

Table 14: Robustness Checks: Long-Short Portfolio Performance

Model	Equal-Weighted				Value-Weighted			
	Pred (%)	Real (%)	SR	PSR	Pred (%)	Real (%)	SR	PSR
Panel A: Winsorisation								
Market (benchmark)	-	0.63	0.14	-	-	0.63	0.14	-
OLS	5.37	2.00	0.32	0.99	4.90	0.92	0.16	0.63
OLS-FF3	0.01	-0.39	-0.11	0.00	0.00	0.18	0.05	0.05
PCR	2.47	1.45	0.20	0.83	2.41	0.75	0.09	0.21
IPCA	4.14	1.47	0.22	0.93	3.84	0.42	0.06	0.09
PLS	4.11	1.81	0.30	0.99	3.87	0.89	0.13	0.47
Linear-En	2.85	2.07	0.35	0.99	2.66	0.97	0.15	0.60
GBRT	5.50	1.83	0.33	0.99	4.71	1.68	0.23	0.94
RF	2.92	1.96	0.31	0.99	2.23	1.40	0.17	0.70
FFNN1	7.10	2.36	0.39	0.99	5.69	1.86	0.23	0.97

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Model	Equal-Weighted				Value-Weighted			
	Pred (%)	Real (%)	SR	PSR	Pred (%)	Real (%)	SR	PSR
FFNN2	6.87	2.24	0.37	0.99	5.48	1.70	0.22	0.94
FFNN3	6.52	2.09	0.34	0.99	5.37	1.43	0.19	0.85
FFNN4	5.79	1.98	0.32	0.99	4.88	1.29	0.18	0.79
FFNN5	4.01	2.02	0.32	0.99	3.47	1.47	0.22	0.95
Non-Linear-En	5.19	2.34	0.37	0.99	4.25	1.68	0.21	0.93
Panel B: Alternative Characteristic Database								
Market (benchmark)	-	0.63	0.14	-	-	0.63	0.14	-
OLS	2.58	0.67	0.12	0.36	2.45	0.66	0.13	0.44
OLS-FF3	0.01	-0.50	-0.14	0.00	0.00	0.20	0.05	0.07
PCR	1.47	0.48	0.10	0.28	1.40	0.11	0.02	0.03
IPCA	2.07	0.83	0.13	0.47	1.96	0.75	0.13	0.46
PLS	2.31	0.83	0.15	0.59	2.17	0.50	0.10	0.24
Linear-En	1.55	0.82	0.14	0.51	1.46	0.40	0.08	0.15
GBRT	1.79	0.10	0.02	0.02	1.69	0.51	0.09	0.22
RF	1.74	0.91	0.14	0.54	1.46	0.63	0.08	0.18
FFNN1	3.60	0.97	0.21	0.89	3.09	0.85	0.14	0.54
FFNN2	3.28	0.95	0.18	0.78	2.80	0.93	0.13	0.46
FFNN3	2.91	1.08	0.20	0.86	2.55	0.48	0.07	0.12
FFNN4	2.29	0.97	0.18	0.75	2.05	0.60	0.09	0.20
FFNN5	1.37	0.85	0.15	0.58	1.24	0.76	0.12	0.39
Non-Linear-En	2.14	1.15	0.21	0.87	1.85	1.07	0.15	0.57
Panel C: Top Market-Capitalisation Segment								
Market (benchmark)	-	0.63	0.14	-	-	0.63	0.14	-
OLS	2.74	0.82	0.16	0.64	2.63	1.05	0.22	0.93
OLS-FF3	0.00	-0.09	-0.02	0.00	0.00	0.17	0.04	0.04
PCR	1.19	0.63	0.09	0.18	1.16	0.71	0.11	0.28
IPCA	1.27	0.26	0.04	0.04	1.22	0.27	0.04	0.04
PLS	1.79	0.94	0.15	0.57	1.73	1.24	0.21	0.89
Linear-En	1.22	0.89	0.16	0.62	1.17	1.22	0.24	0.96
GBRT	2.51	0.63	0.10	0.26	2.40	0.58	0.10	0.27
RF	1.07	0.66	0.10	0.23	1.01	0.91	0.14	0.51
FFNN1	1.58	-0.22	-0.06	0.00	1.65	-0.18	-0.05	0.00
FFNN2	1.01	-0.14	-0.04	0.00	1.03	-0.21	-0.05	0.00
FFNN3	0.76	-0.11	-0.03	0.00	0.77	-0.32	-0.08	0.00
FFNN4	0.68	0.09	0.04	0.03	0.74	0.26	0.08	0.14
FFNN5	0.44	-0.09	-0.03	0.00	0.48	-0.04	-0.01	0.00
Non-Linear-En	0.66	0.37	0.06	0.08	0.65	0.45	0.08	0.17
Panel D: Bottom Market-Capitalisation Segment								
Market (benchmark)	-	0.63	0.14	-	-	0.63	0.14	-
OLS	9.70	3.05	0.34	0.99	9.69	4.00	0.50	0.99
OLS-FF3	0.04	-1.04	-0.22	0.00	0.01	-0.65	-0.12	0.00
PCR	2.96	1.30	0.20	0.85	2.98	1.25	0.17	0.73
IPCA	9.08	3.24	0.39	0.99	9.07	3.39	0.47	0.99
PLS	6.67	2.40	0.31	0.99	6.67	3.02	0.39	0.99
Linear-En	5.12	3.31	0.39	0.99	5.12	3.72	0.49	0.99
GBRT	8.91	3.29	0.42	0.99	8.67	4.29	0.60	0.99
RF	5.23	3.33	0.35	0.99	5.07	3.55	0.41	0.99
FFNN1	14.70	3.86	0.37	0.99	14.24	4.80	0.53	0.99
FFNN2	12.94	3.53	0.36	0.99	12.48	4.40	0.49	0.99
FFNN3	11.33	4.10	0.42	0.99	11.02	4.89	0.56	0.99
FFNN4	9.65	4.11	0.41	0.99	9.40	4.99	0.57	0.99
FFNN5	6.29	3.60	0.39	0.99	6.15	4.22	0.54	0.99
Non-Linear-En	9.03	4.45	0.45	0.99	8.75	5.12	0.58	0.99
Panel E: Alternative Industry Classification								
Market (benchmark)	-	0.63	0.14	-	-	0.63	0.14	-
OLS	5.35	2.03	0.32	0.99	4.90	0.93	0.15	0.60
OLS-FF3	0.01	-0.41	-0.12	0.00	0.00	0.19	0.05	0.06
PCR	2.62	1.58	0.21	0.88	2.63	0.64	0.08	0.15
IPCA	4.14	1.58	0.24	0.97	3.85	0.57	0.08	0.16
PLS	4.15	1.82	0.31	0.99	3.94	0.90	0.13	0.46
Linear-En	2.89	2.11	0.36	0.99	2.71	0.85	0.13	0.42
GBRT	5.59	1.81	0.34	0.99	4.71	1.37	0.21	0.88
RF	2.91	2.06	0.33	0.99	2.18	1.55	0.19	0.83
FFNN1	7.09	2.36	0.39	0.99	5.67	1.56	0.21	0.90
FFNN2	6.82	2.40	0.40	0.99	5.36	1.74	0.25	0.99
FFNN3	6.42	2.22	0.37	0.99	5.24	1.74	0.26	0.99
FFNN4	5.81	2.20	0.36	0.99	4.89	1.78	0.25	0.99
FFNN5	3.71	2.03	0.34	0.99	3.16	1.32	0.18	0.76
Non-Linear-En	5.15	2.42	0.39	0.99	4.17	1.60	0.22	0.93
Panel F: Alternative Sample Split								
Market (benchmark)	-	0.63	0.14	-	-	0.63	0.14	-
OLS	5.68	2.15	0.34	0.99	5.13	1.06	0.19	0.79
OLS-FF3	0.01	-0.39	-0.11	0.00	0.00	0.18	0.05	0.05
PCR	2.60	1.50	0.20	0.83	2.52	0.73	0.09	0.19
IPCA	4.44	1.55	0.22	0.95	4.09	0.17	0.03	0.02
PLS	4.28	1.95	0.33	0.99	4.01	0.90	0.14	0.49
Linear-En	3.00	2.14	0.35	0.99	2.77	0.96	0.15	0.59
GBRT	5.55	2.17	0.39	0.99	4.74	1.85	0.24	0.96
RF	3.14	2.20	0.35	0.99	2.35	1.74	0.22	0.92
FFNN1	7.52	2.44	0.40	0.99	6.11	1.62	0.23	0.96
FFNN2	7.08	2.52	0.42	0.99	5.59	2.02	0.27	0.99
FFNN3	6.60	2.34	0.39	0.99	5.39	1.88	0.27	0.99
FFNN4	5.71	2.29	0.38	0.99	4.84	1.45	0.21	0.92
FFNN5	3.92	2.35	0.40	0.99	3.35	1.75	0.26	0.99
Non-Linear-En	5.25	2.63	0.43	0.99	4.28	2.00	0.27	0.99

Note: This table reports the performance of long-short decile portfolios over the 26-year out-of-sample test period under multiple robustness settings. For each model, the table reports the average predicted return, the average net realised return, the SR and the PSR, separately for equal-weighted and value-weighted portfolio construction. Predicted and net realised returns are expressed in monthly percentages, while SRs are computed from net realised monthly portfolio returns. Bold SRs indicate portfolios that exceed the monthly SR of the market excess return over the same out-of-sample period, equal to 0.14. The PSRs report the probability that the estimated portfolio SR exceeds the market SR, with probabilities lying in the interval [0,1]. Likewise bold PSRs values indicate statistical significance at the 5% level relative to this benchmark.